Airbnb Project
Development Phase

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Phase 2: Presentation

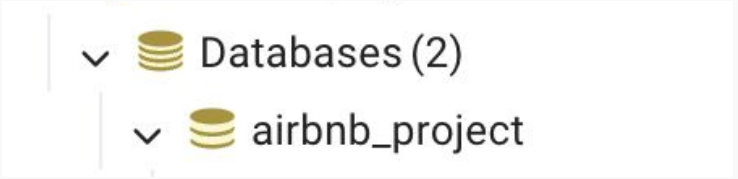
01. Creation of Database

```
1  CREATE DATABASE airbnb_project;
2
```

Data Output | Messages | Notifications

```
CREATE DATABASE

Query returned successfully in 319 msec.
```



A database with name ‘airbnb-project’ was created

02

Creation of Tables

- Creating tables
- Altering tables to add foreign keys
- Insert commands to add into tables

Host:

Create Command:

```
CREATE TABLE IF NOT EXISTS host(
    host_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    location_id character varying(10) NOT NULL,
    payout_id character varying(10) NOT NULL,
    refund_id character varying(10) NOT NULL,
    msg_id character varying(10) NOT NULL,
    id_verify character varying(10) NOT NULL,
    review_id character varying(10) NOT NULL,
    rating_id character varying(10) NOT NULL,
    published_guides character varying(10),
    guide_for_host character varying(10) NOT NULL,
    coupon_id character varying(10) NOT NULL,
    giftcard_id character varying(10) NOT NULL,
    policy_id character varying(10) NOT NULL,
    rules_id character varying(10) NOT NULL,
    cancel_id character varying(10) NOT NULL,
    receipt_id character varying(10) NOT NULL,
    host_email character varying(100) NOT NULL,
    host_password character varying(8) NOT NULL,
    host_name character(100) NOT NULL,
    host_since date NOT NULL,
    host_profile_pic bytea,
    host_phone_no numeric(12) NOT NULL,
    is_super_host boolean NOT NULL,
    host_address character varying(1000) NOT NULL,
    PRIMARY KEY (host_id)
);
```

Changes made in the schema → renamed verify_id to id_verify, host_email character varying (20) changed to character varying (100) to accommodate the email in large capacity

Alter Command:

```
ALTER TABLE IF EXISTS host ADD CONSTRAINT location_id FOREIGN KEY (location_id)
REFERENCES location_id (location_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT review_id FOREIGN KEY (review_id)
REFERENCES review (review_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT rating_id FOREIGN KEY (rating_id)
REFERENCES rating (rating_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT receipt_id FOREIGN KEY (receipt_id)
REFERENCES receipt (receipt_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT msg_id FOREIGN KEY (msg_id)
REFERENCES msg_id (msg_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT payout_id FOREIGN KEY (payout_id)
REFERENCES payouts (payout_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT giftcard_id FOREIGN KEY (giftcard_id)
REFERENCES giftcard (giftcard_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT coupon_id FOREIGN KEY (coupon_id)
REFERENCES coupon (coupon_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT guide_for_host FOREIGN KEY (guide_for_host)
REFERENCES guide_book_for_host (guide_host_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION
NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT published_guide_id FOREIGN KEY (published_guides)
REFERENCES guide_published_by_host (published_guide_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION
NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT policy_id FOREIGN KEY (policy_id)
REFERENCES policy (policy_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT rule_id FOREIGN KEY (rules_id)
REFERENCES rules (rules_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT cancel_id FOREIGN KEY (cancel_id)
REFERENCES cancellation (cancel_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT refund_id FOREIGN KEY (refund_id)
REFERENCES refund (refund_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS host ADD CONSTRAINT id_verify FOREIGN KEY (id_verify)
REFERENCES id_verification (id_verify) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Host : Insert command

- The host table is created for host to have a profile where they can add listing with photos, prices and descriptions.
- They can handle bookings, have guest communication, maintain properties, enhance accommodation experience according to the review and rating which can help them in being superhosts.
- *Changes made in the schema → host_email character varying (20) changed to character varying (100) to accommodate the email in large capacity*
- Below is the insert command for the host table, only the first statement is expanded here as, every insert statement [20 in total] follows a similar structure.

('H001', 'L001', 'LO001', 'PO001', 'RF001', 'MSG00h1', 'IDVF00h1','REID001', 'RAID001', 'PGID001', 'GFH001', 'CC001', 'GC001', 'PO001', 'R001', 'CAN001', 'REC001', 'sara.lindholm@postse.se', 's@lind12', 'Sara Lindholm', '2018-03-11', bytea('/Users/masoomshaikh/Documents/iu-airbnb-project/db_images/host_profiles/host_1.png'), 46701234567, True, 'Drottninggatan 68, 111 21 Stockholm, Sweden')

```
INSERT INTO host
VALUES('H001', 'L001', 'LO001', 'PO001', 'RF001', 'MSG00h1', 'IDVF00h1','REID001', 'RAID001', 'PGID001', 'GFH001', 'CC001', 'GC001', 'PO001', 'R001', 'CAN001'
('H002', 'L002', 'LO002', 'PO002', 'RF002', 'MSG00h2', 'IDVF00h2','REID002', 'RAID002', 'PGID002', 'GFH002', 'CC002', 'GC002', 'PO002', 'R002', 'CAN002', 'REC
('H003', 'L0 03', 'LO003', 'PO003', 'RF003', 'MSG00h3', 'IDVF00h3','REID003', 'RAID003', 'PGID003', 'GFH003', 'CC003', 'GC003', 'PO003', 'R003', 'CAN003', 'RE
('H004', 'L004', 'LO004', 'PO004', 'RF004', 'MSG00h4', 'IDVF00h4','REID004', 'RAID004', 'PGID004', 'GFH004', 'CC004', 'GC004', 'PO004', 'R004', 'CAN004', 'REC
('H005', 'L005', 'LO005', 'PO005', 'RF005', 'MSG00h5', 'IDVF00h5','REID005', 'RAID005', 'PGID005', 'GFH005', 'CC005', 'GC005', 'PO005', 'R005', 'CAN005', 'REC
('H006', 'L006', 'LO006', 'PO006', 'RF006', 'MSG00h6', 'IDVF00h6','REID006', 'RAID006', 'PGID006', 'GFH006', 'CC006', 'GC006', 'PO006', 'R006', 'CAN006', 'REC
('H007', 'L007', 'LO007', 'PO007', 'RF007', 'MSG00h7', 'IDVF00h7','REID007', 'RAID007', 'PGID007', 'GFH007', 'CC007', 'GC007', 'PO007', 'R007', 'CAN007', 'REC
('H008', 'L008', 'LO008', 'PO008', 'RF008', 'MSG00h8', 'IDVF00h8','REID008', 'RAID008', 'PGID008', 'GFH008', 'CC008', 'GC008', 'PO008', 'R008', 'CAN008', 'REC
('H009', 'L009', 'LO009', 'PO009', 'RF009', 'MSG00h9', 'IDVF00h9','REID009', 'RAID009', 'PGID009', 'GFH009', 'CC009', 'GC009', 'PO009', 'R009', 'CAN009', 'REC
('H0010', 'L0010', 'LO0010', 'PO0010', 'RF0010', 'MSG00h10', 'IDVF0h10','REID0010', 'RAID0010', 'PGID0010', 'GFH0010', 'CC0010', 'GC0010', 'PO0010', 'R0010',
('H0011', 'L0011', 'LO0011', 'PO0011', 'RF0011', 'MSG00h11', 'IDVF0h11','REID0011', 'RAID0011', 'PGID0011', 'GFH0011', 'CC0011', 'GC0011', 'PO0011', 'R0011',
('H0012', 'L0012', 'LO0012', 'PO0012', 'RF0012', 'MSG00h12', 'IDVF0h12','REID0012', 'RAID0012', 'PGID0012', 'GFH0012', 'CC0012', 'GC0012', 'PO0012', 'R0012',
('H0013', 'L0013', 'LO0013', 'PO0013', 'RF0013', 'MSG00h13', 'IDVF0h13','REID0013', 'RAID0013', 'PGID0013', 'GFH0013', 'CC0013', 'GC0013', 'PO0013', 'R0013',
('H0014', 'L0014', 'LO0014', 'PO0014', 'RF0014', 'MSG00h14', 'IDVF0h14','REID0014', 'RAID0014', 'PGID0014', 'GFH0014', 'CC0014', 'GC0014', 'PO0014', 'R0014',
('H0015', 'L0015', 'LO0015', 'PO0015', 'RF0015', 'MSG00h15', 'IDVF0h15','REID0015', 'RAID0015', 'PGID0015', 'GFH0015', 'CC0015', 'GC0015', 'PO0015', 'R0015',
('H0016', 'L0016', 'LO0016', 'PO0016', 'RF0016', 'MSG00h16', 'IDVF0h16','REID0016', 'RAID0016', 'PGID0016', 'GFH0016', 'CC0016', 'GC0016', 'PO0016', 'R0016',
('H0017', 'L0017', 'LO0017', 'PO0017', 'RF0017', 'MSG00h17', 'IDVF0h17','REID0017', 'RAID0017', 'PGID0017', 'GFH0017', 'CC0017', 'GC0017', 'PO0017', 'R0017',
('H0018', 'L0018', 'LO0018', 'PO0018', 'RF0018', 'MSG00h18', 'IDVF0h18','REID0018', 'RAID0018', 'PGID0018', 'GFH0018', 'CC0018', 'GC0018', 'PO0018', 'R0018',
('H0019', 'L0019', 'LO0019', 'PO0019', 'RF0019', 'MSG00h19', 'IDVF0h19','REID0019', 'RAID0019', 'PGID0019', 'GFH0019', 'CC0019', 'GC0019', 'PO0019', 'R0019',
('H0020', 'L0020', 'LO0020', 'PO0020', 'RF0020', 'MSG00h20', 'IDVF0h20','REID0020', 'RAID0020', 'PGID0020', 'GFH0020', 'CC0020', 'GC0020', 'PO0020', 'R0020',
```

Guest:

Create Command:

```
CREATE TABLE IF NOT EXISTS guest(
    guest_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    location_id character varying(10) NOT NULL,
    msg_id character varying(10) NOT NULL,
    id_verify character varying(10) NOT NULL,
    rating_id character varying(10) NOT NULL,
    review_id character varying(10) NOT NULL,
    guide_for_guest character varying(10) NOT NULL,
    policy_id character varying(10) NOT NULL,
    rule_id character varying(10) NOT NULL,
    coupon_id character varying(10) NOT NULL,
    giftcard_id character varying(10) NOT NULL,
    refund_id character varying(10) NOT NULL,
    cancel_id character varying(10) NOT NULL,
    payment_id character varying(10) NOT NULL,
    receipt_id character varying(10) NOT NULL,
    guest_name character(100) NOT NULL,
    guest_email character varying(100) NOT NULL,
    guest_password character varying(8) NOT NULL,
    guest_phone_no numeric(12) NOT NULL,
    guest_profile_pic bytea NOT NULL,
    guest_address character varying(100) NOT NULL,
    guest_since date NOT NULL,
    PRIMARY KEY (guest_id)
);
```

Changes made in the schema →renamed verify_id to id_verify, guest_email character varying (20) changed to character varying (100) to accommodate the email in large capacity

Alter Command:

```
ALTER TABLE IF EXISTS guest ADD CONSTRAINT location_id FOREIGN KEY (location_id)
REFERENCES location_id (location_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT review_id FOREIGN KEY (review_id)
REFERENCES review (review_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT rating_id FOREIGN KEY (rating_id)
REFERENCES rating (rating_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT receipt_id FOREIGN KEY (receipt_id)
REFERENCES receipt (receipt_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT booking_id FOREIGN KEY (booking_id)
REFERENCES booking (booking_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT msg_id FOREIGN KEY (msg_id)
REFERENCES msg_id (msg_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT payment_id FOREIGN KEY (payment_id)
REFERENCES payments (payment_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT giftcard_id FOREIGN KEY (giftcard_id)
REFERENCES giftcard (giftcard_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT coupon_id FOREIGN KEY (coupon_id)
REFERENCES coupon (coupon_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT guide_for_guest FOREIGN KEY (guide_for_guest)
REFERENCES guide_book_for_guest (guide_guest_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION
NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT cancel_id FOREIGN KEY (cancel_id)
REFERENCES cancellation (cancel_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT refund_id FOREIGN KEY (refund_id)
REFERENCES refund (refund_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT id_verify FOREIGN KEY (id_verify)
REFERENCES id_verification (id_verify) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT policy_id FOREIGN KEY (policy_id)
REFERENCES policy (policy_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS guest ADD CONSTRAINT rule_id FOREIGN KEY (rules_id)
REFERENCES rules (rules_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Guest : Insert command

- The guest table can be used to find, save to wishlists or book these listings.
- Guests can communicate with hosts using message, leaving feedback in the form of reviews and ratings post-stay.
- They have ability to save suitable payment options and apply them for making payments.
- Guests must obey the mentioned rules and regulations for listings as well as maintain best behaviour.
- Below is the insert command for the guest table, only the first statement is expanded here as, every insert statement [20 in total] follows a similar structure.

('G001', 'L001', 'LO001', 'MSG00g1', 'IDVF00g1', 'RAID001','REID001', 'GFG001', 'P001', 'R001', 'CC001', 'GC001', 'RF001', 'CAN001', 'PY001', 'REC001', 'Ahmed Al-Farsi' , 'ahmed_alFarsi1@postse.se', 'Alfarsi8', 1521234567 , bytea('/Users/masoomshaikh/Documents/iu-airbnb-project/db_images/guest_profiles/guest_1.png'), '23 Sheikh Zayed Road, Dubai, United Arab Emirates','2024-04-14')

```
INSERT INTO guest
VALUES('G001', 'L001', 'LO001', 'MSG00g1', 'IDVF00g1', 'RAID001','REID001', 'GFG001', 'P001', 'R001', 'CC001', 'GC001', 'RF001', 'CAN001', 'PY001', 'REC001'
('G002', 'L002', 'LO002', 'MSG00g2', 'IDVF00g2', 'RAID002','REID002', 'GFG002', 'P002', 'R002', 'CC002', 'GC002', 'RF002', 'CAN002', 'PY002', 'REC002', 'Pr
('G003', 'L003', 'LO003', 'MSG00g3', 'IDVF00g3', 'RAID003','REID003', 'GFG003', 'P003', 'R003', 'CC003', 'GC003', 'RF003', 'CAN003', 'PY003', 'REC003', 'Meh
('G004', 'L004', 'LO004', 'MSG00g4', 'IDVF00g4', 'RAID004','REID004', 'GFG004', 'P004', 'R004', 'CC004', 'GC004', 'RF004', 'CAN004', 'PY004', 'REC004', 'Isa
('G005', 'L005', 'LO005', 'MSG00g5', 'IDVF00g5', 'RAID005','REID005', 'GFG005', 'P005', 'R005', 'CC005', 'GC005', 'RF005', 'CAN005', 'PY005', 'REC005', 'Sar
('G006', 'L006', 'LO006', 'MSG00g6', 'IDVF00g6', 'RAID006','REID006', 'GFG006', 'P006', 'R006', 'CC006', 'GC006', 'RF006', 'CAN006', 'PY006', 'REC006', 'Sop
('G007', 'L007', 'LO007', 'MSG00g7', 'IDVF00g7', 'RAID007','REID007', 'GFG007', 'P007', 'R007', 'CC007', 'GC007', 'RF007', 'CAN007', 'PY007', 'REC007', 'Gra
('G008', 'L008', 'LO008', 'MSG00g8', 'IDVF00g8', 'RAID008','REID008', 'GFG008', 'P008', 'R008', 'CC008', 'GC008', 'RF008', 'CAN008', 'PY008', 'REC008', 'Am
('G009', 'L009', 'LO009', 'MSG00g9', 'IDVF00g9', 'RAID009','REID009', 'GFG009', 'P009', 'R009', 'CC009', 'GC009', 'RF009', 'CAN009', 'PY009', 'REC009', 'Lay
('G0010', 'L0010', 'LO0010', 'MSG00g10', 'IDVF0g10', 'RAID0010','REID0010', 'GFG0010', 'P0010', 'R0010', 'CC0010', 'GC0010', 'RF0010', 'CAN0010', 'PY0010',
('G0011', 'L0011', 'LO0011', 'MSG00g11', 'IDVF0g11', 'RAID0011','REID0011', 'GFG0011', 'P0011', 'R0011', 'CC0011', 'GC0011', 'RF0011', 'CAN0011', 'PY0011',
('G0012', 'L0012', 'LO0012', 'MSG00g12', 'IDVF0g12', 'RAID0012','REID0012', 'GFG0012', 'P0012', 'R0012', 'CC0012', 'GC0012', 'RF0012', 'CAN0012', 'PY0012',
('G0013', 'L0013', 'LO0013', 'MSG00g13', 'IDVF0g13', 'RAID0013','REID0013', 'GFG0013', 'P0013', 'R0013', 'CC0013', 'GC0013', 'RF0013', 'CAN0013', 'PY0013',
('G0014', 'L0014', 'LO0014', 'MSG00g14', 'IDVF0g14', 'RAID0014','REID0014', 'GFG0014', 'P0014', 'R0014', 'CC0014', 'GC0014', 'RF0014', 'CAN0014', 'PY0014',
('G0015', 'L0015', 'LO0015', 'MSG00g15', 'IDVF0g15', 'RAID0015','REID0015', 'GFG0015', 'P0015', 'R0015', 'CC0015', 'GC0015', 'RF0015', 'CAN0015', 'PY0015',
('G0016', 'L0016', 'LO0016', 'MSG00g16', 'IDVF0g16', 'RAID0016','REID0016', 'GFG0016', 'P0016', 'R0016', 'CC0016', 'GC0016', 'RF0016', 'CAN0016', 'PY0016',
('G0017', 'L0017', 'LO0017', 'MSG00g17', 'IDVF0g17', 'RAID0017','REID0017', 'GFG0017', 'P0017', 'R0017', 'CC0017', 'GC0017', 'RF0017', 'CAN0017', 'PY0017',
('G0018', 'L0018', 'LO0018', 'MSG00g18', 'IDVF0g18', 'RAID0018','REID0018', 'GFG0018', 'P0018', 'R0018', 'CC0018', 'GC0018', 'RF0018', 'CAN0018', 'PY0018',
('G0019', 'L0019', 'LO0019', 'MSG00g19', 'IDVF0g19', 'RAID0019','REID0019', 'GFG0019', 'P0019', 'R0019', 'CC0019', 'GC0019', 'RF0019', 'CAN0019', 'PY0019',
('G0020', 'L0020', 'LO0020', 'MSG00g20', 'IDVF0g20', 'RAID0020','REID0020', 'GFG0020', 'P0020', 'R0020', 'CC0020', 'GC0020', 'RF0020', 'CAN0020', 'PY0020',
```

Listing:

Create Command:

```
CREATE TABLE IF NOT EXISTS listing(
    listing_id character varying(10) NOT NULL,
    host_id character varying(10) NOT NULL,
    rules_id character varying(10) NOT NULL,
    location_id character varying(10) NOT NULL,
    rating_id character varying(10) NOT NULL,
    policy_id character varying(10) NOT NULL,
    review_id character varying(10) NOT NULL,
    listing_avail character varying NOT NULL,
    amenities_id character varying(10) NOT NULL,
    listing_name character(100) NOT NULL,
    listing_pic1 bytea NOT NULL,
    listing_pic2 bytea NOT NULL,
    listing_pic3 bytea NOT NULL,
    listing_pic4 bytea NOT NULL,
    listing_description text NOT NULL,
    price_per_night_in_euro double precision NOT NULL,
    listing_address character varying(100) NOT NULL,
    posted_date date NOT NULL,
    is_guest_favourite boolean NOT NULL,
    is_security_deposit boolean NOT NULL,
    security_deposit_amt double precision,
    PRIMARY KEY (listing_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS listing ADD CONSTRAINT host_id FOREIGN KEY (host_id)
REFERENCES host (host_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;

ALTER TABLE IF EXISTS listing ADD CONSTRAINT rule_id FOREIGN KEY (rules_id)
REFERENCES rules (rules_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;

ALTER TABLE IF EXISTS listing ADD CONSTRAINT location_id FOREIGN KEY (location_id)
REFERENCES location (location_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;

ALTER TABLE IF EXISTS listing ADD CONSTRAINT rating_id FOREIGN KEY (rating_id)
REFERENCES rating (rating_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;

ALTER TABLE IF EXISTS listing ADD CONSTRAINT policy_id FOREIGN KEY (policy_id)
REFERENCES policy (policy_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;

ALTER TABLE IF EXISTS listing ADD CONSTRAINT review_id FOREIGN KEY (review_id)
REFERENCES review (review_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;

ALTER TABLE IF EXISTS listing ADD CONSTRAINT listing_avail_id FOREIGN KEY (listing_avail)
REFERENCES listing_avail (listing_avail_id) MATCH SIMPLE ON UPDATE NO ACTION
ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS listing ADD CONSTRAINT amenities_id FOREIGN KEY (amenities_id)
REFERENCES amenities (amenities_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;
```

Changes made in the schema → listing_pic1, listing_pic2, listing_pic3, listing_pic4 added. Renamed price_per_night to price_per_night_in_euro. Changed listing_description bytea to listing_description text for easier output readability

Listing : Insert command

- The listing table is a core component that stores detailed information about each property available for booking.
- Below is the insert command for the listing table, only the first statement is expanded here as, every insert statement [20 in total] follows a similar structure.

('L001', 'H001', 'R001', 'LO001', 'RAID001', 'P001', 'REID001', 'LA001', 'AMID001', 'CouplesPod House Portfolio' , bytea('/Users/masoomshaikh/Documents/iu-airbnb-project/db_images/listing_images/L_1/l1_img1.png'), bytea('/Users/masoomshaikh/Documents/iu-airbnb-project/db_images/listing_images/L_1/l1_img2.png'),bytea('/Users/masoomshaikh/Documents/iu-airbnb-project/db_images/listing_images/L_1/l1_img3.png'),bytea('/Users/masoomshaikh/Documents/iu-airbnb-project/db_images/listing_images/L_1/l1_img4.png'),'If views, luxury, and romantic charm are your thing, we''ve got you covered! Couplespod at The Riverstone House Portfolio is our high end couples retreat.', 547, 'Woleseley, Western Cape, South Africa', '2021-09-14', False, False, null)

```
INSERT INTO listing
VALUES('L001', 'H001', 'R001', 'LO001', 'RAID001', 'P001', 'REID001', 'LA001', 'AMID001',  'CouplesPod House Portfolio' , bytea('/Users/masoomshaikh/Documer
('L002', 'H002', 'R002', 'LO002', 'RAID002', 'P002', 'REID002', 'LA002', 'AMID002',  'Emotional Healing House' , bytea('/Users/masoomshaikh/Documents/iu-air
('L003', 'H003', 'R003', 'LO003', 'RAID003', 'P003', 'REID003', 'LA003', 'AMID003',  'Living Greenhouse Off the Grid' , bytea('/Users/masoomshaikh/Documents
('L004', 'H004', 'R004', 'LO004', 'RAID004', 'P004', 'REID004', 'LA004', 'AMID004',  'Relaxation in Airloft' , bytea('/Users/masoomshaikh/Documents/iu-airbr
('L005', 'H005', 'R005', 'LO005', 'RAID005', 'P005', 'REID005', 'LA005', 'AMID005',  'WonderINN Mirrored Glass Cabin' , bytea('/Users/masoomshaikh/Documents
('L006', 'H006', 'R006', 'LO006', 'RAID006', 'P006', 'REID006', 'LA006', 'AMID006',  'Anemoscope Windmill Villa' , bytea('/Users/masoomshaikh/Documents/iu-a
('L007', 'H007', 'R007', 'LO007', 'RAID007', 'P007', 'REID007', 'LA007', 'AMID007',  'Hornegg Castle: Entire Unit' , bytea('/Users/masoomshaikh/Documents/iu
('L008', 'H008', 'R008', 'LO008', 'RAID008', 'P008', 'REID008', 'LA008', 'AMID008',  'Cottage: Bed & Breakfast by the River' , bytea('/Users/masoomshaikh/Do
('L009', 'H009', 'R009', 'LO009', 'RAID009', 'P009', 'REID009', 'LA009', 'AMID009',  'Apartment Rental' , bytea('/Users/masoomshaikh/Documents/iu-airbnb-pro
('L0010', 'H0010', 'R0010', 'LO0010', 'RAID0010', 'P0010', 'REID0010', 'LA0010', 'AMID0010',  'Dome Stays' , bytea('/Users/masoomshaikh/Documents/iu-airbnb-
('L0011', 'H0011', 'R0011', 'LO0011', 'RAID0011', 'P0011', 'REID0011', 'LA0011', 'AMID0011',  'Leafy Green Earthen Home' , bytea('/Users/masoomshaikh/Docume
('L0012', 'H0012', 'R0012', 'LO0012', 'RAID0012', 'P0012', 'REID0012', 'LA0012', 'AMID0012',  'Glass Igloo Stays' , bytea('/Users/masoomshaikh/Documents/iu-
('L0013', 'H0013', 'R0013', 'LO0013', 'RAID0013', 'P0013', 'REID0013', 'LA0013', 'AMID0013',  'Desert EnSuite Double Tents' , bytea('/Users/masoomshaikh/Doc
('L0014', 'H0014', 'R0014', 'LO0014', 'RAID0014', 'P0014', 'REID0014', 'LA0014', 'AMID0014',  'Cruise Stays' , bytea('/Users/masoomshaikh/Documents/iu-airbr
('L0015', 'H0015', 'R0015', 'LO0015', 'RAID0015', 'P0015', 'REID0015', 'LA0015', 'AMID0015',  'Cave House' , bytea('/Users/masoomshaikh/Documents/iu-airbnb-
('L0016', 'H0016', 'R0016', 'LO0016', 'RAID0016', 'P0016', 'REID0016', 'LA0016', 'AMID0016',  'Tiny Home' , bytea('/Users/masoomshaikh/Documents/iu-airbnb-p
('L0017', 'H0017', 'R0017', 'LO0017', 'RAID0017', 'P0017', 'REID0017', 'LA0017', 'AMID0017',  'Naniwa Ryokan' , bytea('/Users/masoomshaikh/Documents/iu-airb
('L0018', 'H0018', 'R0018', 'LO0018', 'RAID0018', 'P0018', 'REID0018', 'LA0018', 'AMID0018',  'TreeHouse Casarampi' , bytea('/Users/masoomshaikh/Documents/
('L0019', 'H0019', 'R0019', 'LO0019', 'RAID0019', 'P0019', 'REID0019', 'LA0019', 'AMID0019',  'Island Villa: Entire Unit' , bytea('/Users/masoomshaikh/Docum
('L0020', 'H0020', 'R0020', 'LO0020', 'RAID0020', 'P0020', 'REID0020', 'LA0020', 'AMID0020',  'Historical Room Stay' , bytea('/Users/masoomshaikh/Documents/
```

Amenities

Create Command:

```
CREATE TABLE IF NOT EXISTS amenities(
    listing_id character varying(10) NOT NULL,
    amenities_id character varying(10) NOT NULL,
    listing_category_id text NOT NULL,
    amenities_category_id text NOT NULL,
    listing_avail_id character varying(50) NOT NULL,
    PRIMARY KEY (amenities_id)
);
```

Changes made in the schema → amenities_name and availability removed. New cols added: Listing_category_id text, amenities_category_id text, listing_avail_id character varying(50).

Insert Command:

```
ALTER TABLE IF EXISTS amenities
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS amenities
ADD CONSTRAINT listing_category_id FOREIGN KEY (listing_category_id)
REFERENCES listing_category (listing_category_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;


ALTER TABLE IF EXISTS amenities
ADD CONSTRAINT amenities_category_id FOREIGN KEY (amenities_category_id)
REFERENCES amenities_categories (amenities_category_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;


ALTER TABLE IF EXISTS amenities
ADD FOREIGN KEY (listing_avail_id)
REFERENCES listing_avail (listing_avail_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Amenities: Insert Command

- The amenities table stores features and facilities that a listing offers to guests. which can used for filtering search results and enhancing guest experience.
- Below is the insert command for the amenities table, only the first statement is expanded here as, every insert statement [20 in total] follows a similar structure

('L001','AMID001','AID001,AID002,AID003,AID0010,AID0011','ACID001-1,ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID002-1ACID002-2,ACID002-3,ACID002-4,ACID002-5,ACID003-1,ACID004-1,ACID004-2,ACID004-3,ACID004-4,ACID005-1,ACID005-2,ACID005-3,ACID006-1,ACID006-2,ACID007-1,ACID008-1,ACID008-2,ACID008-3,ACID008-4,ACID008-5,ACID008-6,ACID008-7,ACID008-8,ACID008-9,ACID008-10,ACID008-11,ACID008-12,ACID008-13,ACID008-14,ACID008-15,ACID008-16,ACID008-17,ACID008-18,ACID007-2,ACID0010-1,ACID0011-1,ACID0011-2,ACID0011-3,ACID0011-4,ACID0011-5,ACID0010-2','LA001'),

```
INSERT INTO amenities
VALUES('L001','AMID001','AID001,AID002,AID003,AID0010,AID0011','ACID001-1,ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID002-1ACID002-2,ACID0
('L002','AMID002','AID001,AID004','ACID001-1,ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID001-8,ACID001-9,ACID002-1,ACID002-2,ACID002-3,ACI
('L003','AMID003', 'AID001,AID004,AID0018,AID0019,AID0022','ACID001-3,ACID001-7,ACID002-1,ACID002-3,ACID002-4,ACID003-3,ACID004-1,ACID004-2,ACID004-3,ACID004-
('L004','AMID004', 'AID009,AID0014,AID0018,AID0022','ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID001-8,ACID002-1,ACID002-2,ACID002-3,ACID0
('L005','AMID005', 'AID002,AID006,AID0019,AID0020,AID0022','ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID001-8,ACID002-1,ACID002-2,ACID002-
('L006','AMID006', 'AID002,AID006,AID007,AID0011,AID0019,AID0021,AID0022','ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID001-8,ACID002-1,ACI
('L007','AMID007', 'AID001,AID002,AID006,AID008,AID009,AID0011,AID0014,AID0018,AID0021,AID0022','ACID001-2,ACID002-4,ACID002-6,ACID004-1,ACID004-2,ACID004-3,A
('L008','AMID008', 'AID003,AID005,AID0010,AID0011,AID0018,AID0019,AID0020,AID0022','ACID001-1,ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID
('L009','AMID009', 'AID0016,AID0019,AID0022','ACID001-2,ACID001-4,ACID001-7,ACID001-8,ACID002-1,ACID002-2,ACID002-3,ACID002-4,ACID002-6,ACID002-7,ACID003-1,AC
('L0010','AMID0010','AID002,AID005,AID008,AID0010,AID0011,AID0018,AID0019,AID0020','ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID001-8,ACID
('L0011','AMID0011', 'AID0010,AID0014,AID0019,AID0022','ACID001-2,ACID001-4,ACID001-7,ACID002-1,ACID002-2,ACID004-3,ACID005-4,ACID006-1,ACID006-2,ACID007-1,AC
('L0012','AMID0012', 'AID0014,AID008,AID0012,AID0017,AID0020,AID0022','ACID001-2,ACID001-3,ACID001-4,ACID001-7,ACID001-8,ACID002-1,ACID002-2,ACID002-3,ACID002
('L0013','AMID0013', 'AID0013,AID0014,AID0017,AID0020','ACID001-7,ACID002-1,ACID002-3,ACID002-7,ACID004-1,ACID004-2,ACID004-3,ACID004-4,ACID006-2,ACID0010-1,A
('L0014','AMID0014', 'AID003,AID0015,AID0017,AID0018,AID0020','ACID001-1,ACID003-2,ACID004-1,ACID004-2,ACID004-3,ACID004-4,ACID005-3,ACID005-4,ACID006-1,ACID0
('L0015','AMID0015', 'AID002,AID005,AID0018,AID0020,AID0022','ACID001-1,ACID001-2,ACID001-4,ACID001-7,ACID002-1,ACID002-2,ACID002-4,ACID003-2,ACID004-1,ACID00
('L0016','AMID0016', 'AID002,AID0010,AID0011,AID0018,AID0019,AID0020,AID0022','ACID001-1,ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID001-8
('L0017','AMID0017', 'AID0018,AID0019,AID0020','ACID001-2,ACID001-4,ACID001-7,ACID002-6,ACID002-1,ACID002-2,ACID002-3,ACID003-2,ACID005-3,ACID005-4,ACID006-1,
('L0018','AMID0018', 'AID009,AID0010,AID0011,AID0014,AID0018,AID0019','ACID001-2,ACID001-5,ACID001-6,ACID001-7,ACID001-8,ACID001-9,ACID002-1,ACID003-1,ACID003
('L0019','AMID0019', 'AID006,AID0012,AID0014,AID0015,AID0017,AID0018,AID0022','ACID001-3,ACID001-7,ACID001-9,ACID002-1ACID002-3,ACID002-4,ACID002-5,ACID003-1,
('L0020','AMID0020', 'AID0016,AID0018,AID0022','ACID001-2,ACID001-3,ACID001-4,ACID001-5,ACID001-6,ACID001-7,ACID001-8,ACID001-9,ACID002-1,ACID002-2,ACID002-3,
```

Amenities Category

Create Command:

```
CREATE TABLE IF NOT EXISTS amenities_categories(
    amenities_category_id character varying(10) NOT NULL,
    amenities_category_name character varying(10) NOT NULL,
    amenities_name text NOT NULL
);
```

Insert Command:

```
INSERT INTO amenities_category
VALUES('ACID001-1','Bathroom','Bath'),
('ACID001-2','Bathroom','Hair Dryer'),
('ACID001-3','Bathroom','Cleaning Products'),
('ACID001-4','Bathroom','Shampoo'),
('ACID001-5','Bathroom','Conditioner'),
('ACID001-6','Bathroom','Body Soap'),
('ACID001-7','Bathroom','Hot Water'),
('ACID001-8','Bathroom','Shower Gel'),
('ACID001-9','Bathroom','Bidet'),
('ACID002-1','Bedroom & Laundry','Essentials(Towels,
Bedsheets, Soap and Toilet Paper)'),
('ACID002-2','Bedroom','Hangers'),
('ACID002-3','Bedroom','Bed Linen'),
('ACID002-4','Bedroom','Iron'),
('ACID002-5','Bedroom','Clothes Storage: Wardrobe'),
('ACID002-6','Bedroom','Free Washing Machine'),
('ACID002-7','Bedroom','Extra Pillows and Blankets'),
('ACID002-8','Bedroom','Room-Darkening Blinds'),
('ACID002-9','Bedroom','Mosquito Net'),
('ACID003-1','Entertainment','Books and Reading Materials'),
('ACID003-2','Entertainment','HDTV with Netflix'),
('ACID003-3','Entertainment','Pocket Wifi'),
('ACID003-3','Entertainment','Ethernet Connection'),
('ACID003-4','Entertainment','Sound System'),
('ACID003-5','Entertainment','Games Console'),
('ACID003-6','Entertainment','Exercise Equipment'),
('ACID003-7','Entertainment','Board Games'),
('ACID004-1','Family','Cot'),
```

Changes made in the schema →
new table amenities_category created.

- The amenities category table is a categorised list of all the possible amenities a listing or property has to offer to a guest.
- Each amenity[eg. Bathroom] has its own category-arranged attributes[eg. Hair-Dryer] and
- Here is the insert command for the amenities category table, only the first statement is expanded here as, every insert statement[75 in total] follows a similar structure.

('ACID001-1','Bathroom','Bath')

Listing Avail:

Create Command:

```
CREATE TABLE IF NOT EXISTS listing_avail(
    listing_avail_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    no_of_beds numeric(10) NOT NULL,
    no_of_bath numeric(10) NOT NULL,
    max_no_of_guest numeric(50) NOT NULL,
    guest_type character varying(50) NOT NULL,
    is_pets_allowed boolean NOT NULL,
    no_of_pets character varying(5),
    PRIMARY KEY (listing_avail_id)
);
```

Changes made in the schema → check_in and check_out removed. Renamed no_of_guest to max_no_of_guest

Alter Command:

```
ALTER TABLE IF EXISTS listing_avail
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID;
```

Insert Command:

```
INSERT INTO listing_avail
VALUES('LA001','L001',1,1,2,'Adults Only',True,2),
('LA002','L002',1,1,2,'Adults Only',False,0),
('LA003','L003',4,2,6,'All Ages Allowed',True,3),
('LA004','L004',2,1,2,'Adults Only',False,0),
('LA005','L005',6,3,8,'All Ages Allowed',True,3),
('LA006','L006',6,3,8,'Adults Only',False,0),
('LA007','L007',20,10,25,'All Ages Allowed',True,9),
('LA008','L008',6,3,8,'Adults Only',True,3),
('LA009','L009',3,2,4,'All Ages Allowed',False,0),
('LA0010','L0010',3,2,4,'All Ages Allowed',False,0),
('LA0011','L0011',8,3,9,'All Ages Allowed',True,3),
('LA0012','L0012',8,4,8,'All Ages Allowed',True,4),
('LA0013','L0013',8,3,8,'All Ages Allowed',True,4),
('LA0014','L0014',15,6,15,'All Ages Allowed',True,6),
('LA0015','L0015',3,2,4,'All Ages Allowed',False,0),
('LA0016','L0016',2,1,2,'Adults Only',True,2),
('LA0017','L0017',6,3,8,'Adults Only',False,0),
('LA0018','L0018',3,2,5,'All Ages Allowed',True,2),
('LA0019','L0019',8,3,9,'All Ages Allowed',True,3),
('LA0020','L0020',2,1,2,'Adults Only',True,1);
```

- The listing avail table shows the data or availability of a listing through which it can be booked by a guest.
- The details such as maximum number of guests, age category and many other features can be stored in this table.
- Here is the insert command for the listing avail table, [20 in total]:

Listing Category

Create Command:

```
CREATE TABLE IF NOT EXISTS listing_category(
    listing_category_id character varying(10) NOT NULL,
    category_name character(100) NOT NULL,
    PRIMARY KEY (listing_category_id)
);
```

- The listing category is a classification system that applies to listing in a group.
- This classification is based on its features such as style of the listing, the view it offers or the structure of the listing
- Here is the insert command for the listing category table, [22 in total]:

Insert Command:

```
INSERT INTO listing_category
VALUES ('AID001', 'Lake View'),
('AID002', 'Mountain View'),
('AID003', 'Waterfront'),
('AID004', 'Shared Beach Access'),
('AID005', 'River View'),
('AID006', 'Sea View'),
('AID007', 'Bay View'),
('AID008', 'Lake Access'),
('AID009', 'Park View'),
('AID0010', 'Garden View'),
('AID0011', 'Valley View'),
('AID0012', 'Beach Access- Beachfront'),
('AID0013', 'Sand Dunes View'),
('AID0014', 'Garden'),
('AID0015', 'Ocean View'),
('AID0016', 'City Skyline View'),
('AID0017', 'Star Gazing'),
('AID0018', 'Private Patio or Balcony'),
('AID0019', 'Private Back Garden'),
('AID0020', 'Indoor Fireplace'),
('AID0021', 'Vineyard View'),
('AID0022', 'Kitchen');
```

Admin:

Create Command:

```
CREATE TABLE IF NOT EXISTS admin(
    admin_id character varying(10) NOT NULL,
    host_id character varying(10) NOT NULL,
    guest_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    id_verify character varying(10) NOT NULL,
    review_id character varying(10) NOT NULL,
    rating_id character varying(10) NOT NULL,
    receipt_id character varying(10) NOT NULL,
    booking_id character varying(10) NOT NULL,
    listing_avail character varying(10) NOT NULL,
    payout_id character varying(10) NOT NULL,
    payment_id character varying(10) NOT NULL,
    guide_for_host character varying(10) NOT NULL,
    guide_by_host character varying(10) NOT NULL,
    guide_for_guest character varying(10) NOT NULL,
    giftcard_id character varying(10) NOT NULL,
    coupon_id character varying(10) NOT NULL,
    cancel_id character varying(10) NOT NULL,
    refund_id character varying(10) NOT NULL,
    admin_email character varying(20) NOT NULL,
    admin_password character varying(8) NOT NULL,
    admin_name character(20) NOT NULL,
    admin_profile_pic bytea NOT NULL,
    designation character(100) NOT NULL,
    admin_address character varying(100) NOT NULL,
    admin_phone_no numeric(12) NOT NULL,
    joined_date date NOT NULL,
    PRIMARY KEY (admin_id)
);
```

Changes made in the schema → region_id, country_id, city_id, Neighborhood_id removed

Alter Command:

```
ALTER TABLE IF EXISTS admin ADD CONSTRAINT host_id FOREIGN KEY (host_id)
REFERENCES host (host_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT review_id FOREIGN KEY (review_id)
REFERENCES review (review_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT rating_id FOREIGN KEY (rating_id)
REFERENCES rating (rating_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT receipt_id FOREIGN KEY (receipt_id)
REFERENCES receipt (receipt_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT booking_id FOREIGN KEY (booking_id)
REFERENCES booking (booking_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT listing_avail_id FOREIGN KEY (listing_avail)
REFERENCES listing_avail (listing_avail_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION
NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT payout_id FOREIGN KEY (payout_id)
REFERENCES payouts (payout_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT payment_id FOREIGN KEY (payment_id)
REFERENCES payments (payment_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT guide_by_host FOREIGN KEY (guide_by_host)
REFERENCES guide_published_by_host (published_guide_id) MATCH SIMPLEON UPDATE NO ACTION ON DELETE
NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT guide_for_host FOREIGN KEY (guide_for_host)
REFERENCES guide_book_for_host (guide_host_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION
NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT giftcard_id FOREIGN KEY (giftcard_id)
REFERENCES giftcard (giftcard_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT coupon_id FOREIGN KEY (coupon_id)
REFERENCES coupon (coupon_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT guide_for_guest FOREIGN KEY (guide_for_guest)
REFERENCES guide_book_for_guest (guide_guest_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION
NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT cancel_id FOREIGN KEY (cancel_id)
REFERENCES cancellation (cancel_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT refund_id FOREIGN KEY (refund_id)
REFERENCES refund (refund_id) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS admin ADD CONSTRAINT id_verify FOREIGN KEY (id_verify)
REFERENCES id_verification (id_verify) MATCH SIMPLE ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Admin: Insert command

- The admin table is responsible for all the internal operations such as managing platform activity, enforcing compliance with community standards and legal specifications, resolving conflicts like refunds, penalties and cancellations.
- Below is the insert command for the admin table, only the first statement is expanded here as, every insert statement [20 in total] follows a similar structure.

('AD001','H001','G001','L001', 'IDVF00a1', 'REID001', 'RAID001', 'REC001', 'BO001', 'LA001', 'PO001', 'PY001', 'GFH001', 'PGID001', 'GFG001', 'GC001','CC001','CAN001','RF001', 'antonio.rossi@postamail.it','Anton!0','Antonio Rossi', bytea('/Users/masoomshaikh/Documents/iu-airbnb-project/db_images/admin_profiles/admin_1.png'),'Website Manager', 'Via Roma 34, 00184 Rome, Italy','3934567890','2016-07-11')

```
INSERT INTO admin
VALUES('AD001','H001','G001','L001', 'IDVF00a1','REID001','RAID001','REC001','BO001','LA001','PO001','PY001','GFH001','PGID001','GFG001','GC001','CC001','CA
('AD002','H002','G002','L002', 'IDVF00a2','REID002','RAID002','REC002','BO002','LA002','PO002','PY002','GFH002','PGID002','GFG002','GC002','CC002','CAN002',
('AD003','H003','G003','L003', 'IDVF00a3','REID003','RAID003','REC003','BO003','LA003','PO003','PY003','GFH003','PGID003','GFG003','GC003','CC003','CAN003',
('AD004','H004','G004','L004', 'IDVF00a4','REID004','RAID004','REC004','BO004','LA004','PO004','PY004','GFH004','PGID004','GFG004','GC004','CC004','CAN004',
('AD005','H005','G005','L005', 'IDVF00a5','REID005','RAID005','REC005','BO005','LA005','PO005','PY005','GFH005','PGID005','GFG005','GC005','CC005','CAN005',
('AD006','H006','G006','L006', 'IDVF00a6','REID006','RAID006','REC006','BO006','LA006','PO006','PY006','GFH006','PGID006','GFG006','GC006','CC006','CAN006',
('AD007','H007','G007','L007', 'IDVF00a7','REID007','RAID007','REC007','BO007','LA007','PO007','PY007','GFH007','PGID007','GFG007','GC007','CC007','CAN007',
('AD008','H008','G008','L008', 'IDVF00a8','REID008','RAID008','REC008','BO008','LA008','PO008','PY008','GFH008','PGID008','GFG008','GC008','CC008','CAN008',
('AD009','H009','G009','L009', 'IDVF00a9','REID009','RAID009','REC009','BO009','LA009','PO009','PY009','GFH009','PGID009','GFG009','GC009','CC009','CAN009',
('AD0010','H0010','G0010','L0010', 'IDVF00a10','REID0010','RAID0010','REC0010','BO0010','LA0010','PO0010','PY0010','GFH0010','PGID0010','GFG0010','GC0010','
('AD0011','H0011','G0011','L0011', 'IDVF00a11','REID0011','RAID0011','REC0011','BO0011','LA0011','PO0011','PY0011','GFH0011','PGID0011','GFG0011','GC0011','
('AD0012','H0012','G0012','L0012', 'IDVF00a12','REID0012','RAID0012','REC0012','BO0012','LA0012','PO0012','PY0012','GFH0012','PGID0012','GFG0012','GC0012','
('AD0013','H0013','G0013','L0013', 'IDVF00a13','REID0013','RAID0013','REC0013','BO0013','LA0013','PO0013','PY0013','GFH0013','PGID0013','GFG0013','GC0013','
('AD0014','H0014','G0014','L0014', 'IDVF00a14','REID0014','RAID0014','REC0014','BO0014','LA0014','PO0014','PY0014','GFH0014','PGID0014','GFG0014','GC0014','
('AD0015','H0015','G0015','L0015', 'IDVF00a15','REID0015','RAID0015','REC0015','BO0015','LA0015','PO0015','PY0015','GFH0015','PGID0015','GFG0015','GC0015','
('AD0016','H0016','G0016','L0016', 'IDVF00a16','REID0016','RAID0016','REC0016','BO0016','LA0016','PO0016','PY0016','GFH0016','PGID0016','GFG0016','GC0016','
('AD0017','H0017','G0017','L0017', 'IDVF00a17','REID0017','RAID0017','REC0017','BO0017','LA0017','PO0017','PY0017','GFH0017','PGID0017','GFG0017','GC0017','
('AD0018','H0018','G0018','L0018', 'IDVF00a18','REID0018','RAID0018','REC0018','BO0018','LA0018','PO0018','PY0018','GFH0018','PGID0018','GFG0018','GC0018','
('AD0019','H0019','G0019','L0019', 'IDVF00a19','REID0019','RAID0019','REC0019','BO0019','LA0019','PO0019','PY0019','GFH0019','PGID0019','GFG0019','GC0019','
('AD0020','H0020','G0020','L0020', 'IDVF00a20','REID0020','RAID0020','REC0020','BO0020','LA0020','PO0020','PY0020','GFH0020','PGID0020','GFG0020','GC0020','
```

Receipt:

Create Command:

```
CREATE TABLE IF NOT EXISTS receipt(
    receipt_id character varying(10) NOT NULL,
    coupon_id character varying(10),
    giftcard_id character varying(10),
    booking_id character varying(10) NOT NULL,
    receipt_generation_date date NOT NULL,
    security_deposit double precision NOT NULL,
    booking_amt double precision NOT NULL,
    coupon_discount character varying(50) NOT NULL,
    giftcard_discount character varying(50) NOT NULL,
    cleaning_fee double precision NOT NULL,
    taxes_added double precision NOT NULL,
    total_amt double precision NOT NULL,
    discounted_amt double precision NOT NULL,
    security_deposit_status character varying(100) NOT NULL,
    booking_status character varying(100) NOT NULL,
    PRIMARY KEY (receipt_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS receipt
ADD CONSTRAINT giftcard_id FOREIGN KEY (giftcard_id)
REFERENCES giftcard (giftcard_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS receipt
ADD CONSTRAINT booking_id FOREIGN KEY (booking_id)
REFERENCES booking (booking_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS wishlists
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Changes made in the schema →

New cols added: booking_amt, coupon_discount ,giftcard_discount, discounted_amt

Receipt: Insert command

- The receipt table is a record that confirm the financial transaction and booking for a listing was done by a guest.
- Receipt table consists of fields such as payment details, guest details and many more which can be verified by the guest and host while checking in or handling disputes such as cancellations or refunds
- Below is the insert command for the receipt table, only the first statement is expanded here as, every insert statement [40 in total] follows a similar structure.

('REC001','CC001','GC0010','BO001','2021-11-26',0.00,1592,'10%','0',50,90,1732,1558,'Not Applicable','Confirmed')

```
INSERT INTO receipt
VALUES('REC001','CC001','GC0010','BO001','2021-11-26',0.00,1592,'10%','0',50,90,1732,1558,'Not Applicable','Confirmed'),
('REC002','CC001','GC0016','BO002','2022-10-06',0.00,1641,'10%','0',45,80,1766,1589,'Not Applicable','Cancelled'),
('REC003','CC005','GC0010','BO003','2021-01-02',0.00,1347,'15%','15%',50,100,1497,1047,'Not Applicable','Confirmed'),
('REC004','CC004','GC003','BO004','2021-12-26',500,3675,'0','30%',65,110,4350,3045,'Paid','Confirmed'),
('REC005','CC008','GC0014','BO005','2024-05-30',340,3934,'10%','10%',70,140,4484,3587,'Paid','Cancelled'),
('REC006','CC001','GC0010','BO006','2025-03-21',300,2496,'10%','0',50,100,2946,2651,'Paid','Cancelled'),
('REC007','CC002','GC0010','BO007','2022-06-06',400,1828,'0','35%',35,70,2333,1516,'Paid','Confirmed'),
('REC008','CC009','GC0013','BO008','2024-10-31',200,3930,'35%','0',70,135,4335,2817.75,'Paid','Confirmed'),
('REC009','CC0010','GC008','BO009','2023-07-27',500,4995,'15%','10%',100,150,5745,3734.25,'Paid','Cancelled'),
('REC0010','CC001','GC0016','BO0010','2025-04-28',310,890,'10%','0',35,70,1305,1174.50,'Paid','Confirmed'),
('REC0011','CC0020','GC0015','BO0011','2024-09-23',300,1732,'20%','0',45,90,2167,1733.60,'Paid','Cancelled'),
('REC0012','CC0011','GC007','BO0012','2023-06-27',0.00,1098,'25%','0',40,80,1218,913,'Not Applicable','Confirmed'),
('REC0013','CC0012','GC0010','BO0013','2024-01-19',220,2901,'15%','15%',50,100,3271,2289,'Paid','Cancelled'),
('REC0014','CC001','GC0010','BO0014','2023-09-27',310,1335,'10%','15%',50,90,1785,1338.75,'Paid','Confirmed'),
('REC0015','CC0017','GC008','BO0015','2021-11-26',0.00,2786,'35%','10%',70,105,2961,1628.55,'Not Applicable','Cancelled'),
('REC0016','CC001','GC0013','BO0016','2019-03-13',220,4854,'10%','0',80,140,5294,4764.60,'Paid','Confirmed'),
('REC0017','CC0018','GC0014','BO0017','2019-10-01',0.00,642,'15%','0',50,80,772,656.20,'Not Applicable','Cancelled'),
('REC0018','CC001','GC0015','BO0018','2023-05-27',350,4550,'10%','0',90,145,5035,4531.50,'Paid','Confirmed'),
('REC0019','CC0019','GC001','BO0019','2025-02-28',0.00,682,'15%','',50,80,758,644.30,'Not Applicable','Cancelled'),
('REC0020','CC0016','GC0010','BO0020','2024-11-30',500,547,'0','20%',50,50,1147,917.60,'Paid','Confirmed'),
('REC0021','CC0020','GC001','BO0021','2021-12-24',200,2620,'20%','20%',70,110,3000,1800,'Paid','Cancelled'),
('REC0022','CC0010','GC005','BO0022','2020-10-31',320,1686,'15%','10%',45,105,2156,1401.40,'Paid','Confirmed'),
('REC0023','CC0015','GC0011','BO0023','2024-06-22',340,2496,'0','15%',65,115,3016,2563.60,'Paid','Confirmed'),
('REC0024','CC0018','GC004','BO0024','2021-12-17',500,5994,'15%','35%',105,170,6769,3384.50,'Paid','Cancelled'),
('REC0025','CC0012','GC0013','BO0025','2021-11-26',500,1462,'15%','15%',50,100,2112,1478.40,'Paid','Confirmed'),
('REC0026','CC001','GC003','BO0026','2018-11-25',0.00,1641,'10%','0',50,100,1791,1611.90,'Not Applicable','Cancelled'),
('REC0027','CC0013','GC0010','BO0027','2024-09-26',0.00,3143,'0','15%',85,120,3348,2845.80,'Not Applicable','Confirmed'),
('REC0028','CC0012','GC008','BO0028','2023-09-30',220,3868,'15%','10%',70,105,4231,2808.65,'Paid','Cancelled'),
('REC0029','CC0016','GC0012','BO0029','2023-09-27',600,6120,'0','15%',100,130,6950,5907.50,'Paid','Confirmed'),
```

Booking:

Create Command:

```
CREATE TABLE IF NOT EXISTS booking(
    booking_id character varying(10) NOT NULL,
    guest_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    number_of_guests character varying(200) NOT NULL,
    number_of_pets character varying(20) NOT NULL,
    nightly_price double precision NOT NULL,
    staying_for character varying(20) NOT NULL,
    payable_amt double precision NOT NULL,
    pay_method character varying(20) NOT NULL,
    booking_date date NOT NULL,
    booking_status character varying(100) NOT NULL,
    security_deposit boolean NOT NULL,
    PRIMARY KEY (booking_id)
);
```

Changes made in the schema →
Changed pay_method character varying(10) changed to (20),
New cols added:
number_of_guests character varying(200) NOT NULL,
number_of_pets character varying(20) NOT NULL,
nightly_price double precision NOT NULL,
staying_for character varying(20) NOT NULL,
payable_amt double precision NOT NULL

Alter Command:

```
ALTER TABLE IF EXISTS booking
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS booking
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

- The booking table is a central that is managing the reservations made by various guests.
- This table stores all the details of a booking made by a guest for a specific property.
- Below is the insert command for the booking table, only the first statement is expanded here as, every insert statement [40 in total] follows a similar structure.

('BO001', 'G001', 'L008', '2 Adults', '0', 398, '4 Nights', 1592, 'Credit Card', '2021-11-26', 'Confirmed', False)

Booking: Insert command

```
INSERT INTO booking
VALUES('BO001', 'G001', 'L008', '2 Adults', '0', 398, '4 Nights', 1592, 'Credit Card', '2021-11-26', 'Confirmed', False),
('BO002', 'G0010', 'L001', '2 Adults', '1 Dog', 547, '3 Nights', 1641, 'PayPal', '2022-10-06', 'Cancelled', False),
('BO003', 'G0012', 'L003', '2 Adults, 1 Child', '0', 449, '2 Nights', 1347 ,'Debit Card', '2021-01-02', 'Confirmed', False),
('BO004', 'G0019', 'L0020', '2 Adults', '1 Cat', 731, '5 Nights', 3675,'Booking Credits', '2021-12-26', 'Confirmed', True),
('BO005', 'G0014', 'L0017', '6 Adults', '0', 562, '7 Nights', 3934,'Credit Card', '2024-05-30', 'Cancelled', True),
('BO006', 'G0016', 'L0018', '2 Adults, 2 Children', '0', 624, '3 Nights', 2496,'Apple Pay', '2025-03-21', 'Cancelled', True),
('BO007', 'G0018', 'L004', '1 Adult', '0', 914, '2 Nights', 1828,'PayPal', '2022-06-06', 'Confirmed', True),
('BO008', 'G007', 'L002', '1 Adults', '0', 655, '6 Nights', 3930,'Debit Card', '2024-10-31', 'Confirmed', True),
('BO009', 'G006', 'L0019', '2 Adults', '0', 999, '5 Nights', 4995,'PayPal', '2023-07-27', 'Cancelled', True),
('BO0010', 'G003', 'L0013', '3 Adults', '0', 445, '2 Night', 890, 'Credit Card', '2025-04-28', 'Confirmed', True),
('BO0011', 'G004', 'L006', '2 Adults', '0', 433, '4 Nights', 1732, 'Apple Pay', '2024-09-23', 'Cancelled', True),
('BO0012', 'G002', 'L009', '1 Adult', '0', 549, '2 Nights', 1098, 'Booking Credits', '2023-06-27', 'Confirmed', False),
('BO0013', 'G001', 'L0015', '2 Adults', '0', 967, '3 Nights', 2901, 'PayPal', '2024-01-19', 'Cancelled', True),
('BO0014', 'G0016', 'L0013', '1 Adult', '1 Dog', 445, '3 Nights', 1335, 'PayPal', '2023-09-27', 'Confirmed', True),
('BO0015', 'G006', 'L008', '2 Adults', '2 Cats', 398, '7 Nights', 2786, 'Booking Credits', '2021-11-26', 'Cancelled', False),
('BO0016', 'G009', 'L0010', '3 Adults', '0', 809, '6 Nights', 4854, 'Booking Credits', '2019-03-13', 'Confirmed', True),
('BO0017', 'G0020', 'L0016', '1 Adult', '1 Dog', 321, '2 Nights', 642, 'Credit Card', '2019-10-01', 'Cancelled', False),
('BO0018', 'G009', 'L0012', '5 Adults', '0', 650, '7 Nights', 4550, 'Debit Card', '2023-05-27', 'Confirmed', True),
('BO0019', 'G0013', 'L0011', '2 Adults, 2 Children', '1 Cat, 1 Dog', 341, '2 Nights', 682, 'Apple Pay', '2025-02-28', 'Cancelled', True),
('BO0020', 'G0018', 'L001', '1 Adult', '1 Chameleon', 547, '1 Nights', 547, 'Credit Card', '2024-11-30', 'Confirmed', True),
('BO0021', 'G008', 'L002', '1 Adult', '1 Hamster', 655, '4 Nights', 2620, 'PayPal', '2021-12-24', 'Cancelled', True),
('BO0022', 'G003', 'L0017', '6 Adults', '0', 562, '3 Nights', 1686, 'Credit Card', '2020-10-31', 'Confirmed', True),
('BO0023', 'G002', 'L0018', '2 Adults', '0', 624, '4 Nights', 2496, 'Debit Card', '2024-06-22', 'Confirmed', True),
('BO0024', 'G0019', 'L0019', '3 Adults, 1 Child', '1 Parrot', 999, '6 Nights', 5994, 'Debit Card', '2021-12-17', 'Cancelled', True),
('BO0025', 'G0017', 'L0020', '1 Adult', '0', 731, '2 Nights', 1462, 'Credit Card', '2021-11-26', 'Confirmed', True),
('BO0026', 'G0015', 'L005', '2 Adults', '1 Dog', 547, '3 Nights', 1641, 'Apple Pay', '2018-11-25', 'Cancelled', False),
('BO0027', 'G0020', 'L003', '4 Adults', '2 Parrots, 1 Dog', 449, '7 Nights', 3143, 'Apple Pay', '2024-09-26', 'Confirmed', False),
('BO0028', 'G0016', 'L0015', '2 Adults, 3 Children', '0', 967, '4 Nights', 3868, 'Booking Credits', '2023-09-30', 'Cancelled', True),
('BO0029', 'G0014', 'L007', '20 Adults, 7 Children', '3 Dogs', 612, '10 Nights', 6120, 'Credit Card', '2023-09-27', 'Confirmed', True),
```

Policy

- The policy table stores conditions associated with bookings and listings such as cancellation policy, refund policy, and other terms of stay.

Create Command:

```
CREATE TABLE IF NOT EXISTS policy(
    policy_id character varying(10) NOT NULL,
    policy_name character(100) NOT NULL,
    policy_text text NOT NULL,
    PRIMARY KEY (policy_id)
);
```

Changes made in the schema →
policy_text bytea changed to policy_text text, policy_category deleted

- Below is the insert command for the policy table, only the first statement is expanded here as, every insert statement[13 in total] follows a similar structure.

Insert Command:

```
INSERT INTO policy
VALUES('POID001','Refund for Cancelled Reservations','If you need to cancel a reservation, yo
('POID002','Refund If Cancelled before Check-In','Refunds are subject to your host's cancella
('POID003','Refund If Cancelled after Check-In','Refunds are subject to your host's cancellat
('POID004','Refund Policy-Original Method','Refunds are issued to the original payment metho
('POID005','Cancellation Within 24 Hours of Booking','If you cancel within the first 24 hours
('POID006','Cancellation deadlines and refunds','The times and dates we display for cancellat
('POID007','Paid with Coupon','If you cancel a reservation in which you used a coupon to pay
('POID008','Paid with Booking Credit','Refunds for booking credit are subject to your host's
('POID009','Cancellation Policies for your listing','Your standard cancellation policy applie
('POID0010','If Hosts cancels a booking','Cancelling a guest's reservation can have serious
('POID0011','Cancelling a reservation as a Host','We understand that cancellations are somet
('POID0012','Smoking Policy','If you don''t allow smoking, we suggest posting signs to remind
('POID0013','Behavior Commitment','I agree to treat everyone in the community - regardless o
```

('POID001','Refund for Cancelled Reservations','If you need to cancel a reservation, you may be eligible for a refund – it all depends on your Host's cancellation policy and when you cancel the reservation.'),

Rules

- The rules table stores rules associated to specific listing that defines guest behavior [what guests can and cannot do] during their stay to ensure safety, respect for the property, and a smooth hosting experience.

  Changes made in the schema → rules_text bytea changed to rules_text text, rules_category deleted

Create Command:

```
CREATE TABLE IF NOT EXISTS rules(
    rules_id character varying(10) NOT NULL,
    rules_name character varying(100) NOT NULL,
    rules_text text NOT NULL,
    PRIMARY KEY (rules_id)
);
```

- Below is the insert command for the rules table, only the first statement is expanded here as, every insert statement[7 in total] follows a similar structure.

Insert Command:

('ROO1','Health and Cleanliness Rules','Staying organised can help you make the most of your time. Store all your cleaning supplies in one place, like a portable storage tray or an owner's locked cupboard, and restock regularly.'),

```
INSERT INTO rules
VALUES('ROO1','Health and Cleanliness Rules','Staying organised can help you r
('ROO2','Safety','Indicate local emergency numbers and the nearest hospital. I
('ROO3','Preventive Measures for Fire Hazard','Ensure you have a functioning s
('ROO4','Privacy','Always be mindful of your guests'' privacy. Fully disclose
('ROO5','Access & Child-Proofing','Go through your home to identify any areas
('ROO6','Building Rules','Ensure you relay your building''s common area rules
('ROO7','House Rules','To avoid surprises, you may want to include the informa
```

Cancellation:

Create Command:

```
CREATE TABLE IF NOT EXISTS cancellation(
    cancel_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    guest_id character varying(10) NOT NULL,
    host_id character varying(10),
    refund_id character varying(10) NOT NULL,
    receipt_id character varying(10) NOT NULL,
    cancel_reason character(100) NOT NULL,
    cancel_date date NOT NULL,
    cancel_status character(100) NOT NULL
);
```

- The cancellation table stores all the details about cancellation of a specific booking.
- It helps in processing standard measures for handling applied cancellations.
- It also supports queries regarding cancelling a booking and refund options.

Changes in the schema →
Primary key: removed, rules_id and policy_id removed

Alter Command:

```
ALTER TABLE IF EXISTS cancellation
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS cancellation
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION
NOT VALID;

ALTER TABLE IF EXISTS cancellation
ADD CONSTRAINT host_id FOREIGN KEY (host_id)
REFERENCES host (host_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS cancellation
ADD CONSTRAINT refund_id FOREIGN KEY (refund_id)
REFERENCES refund (refund_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS cancellation
ADD CONSTRAINT receipt FOREIGN KEY (receipt_id)
REFERENCES receipt (receipt_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Cancellation: Insert command

Below is the insert command for the cancellation table, [20 in total]:

```
INSERT INTO cancellation
VALUES('CAN0010','L001','G0010','H001','RF0010','REC002','Payment Failure','2022-10-06','Transaction Failed'),
('CAN0014','L0017','G0014','H0017','RF0014','REC005','Other, Personal Reason','2024-05-31','Applied for Refund'),
('CAN0016','L0018','G0016','H0018','RF0016','REC006','Other, Need to change booking the dates','2025-03-23',
'Help from Customer Service'),
('CAN0019','L0019','G006','H0019','RF006','REC006','Cancellation made by host: Pipe Damages','2023-07-29',
'Providing a Full Refund'),
('CAN004','L006','G004','H006','RF004','REC0011','Payment Failure','2024-09-23','Transaction Failed'),
('CAN001','L0015','G001','H0015','RF001','REC0013','Payment Failure','2024-01-19','Transaction Failed'),
('CAN006','L008','G006','H008','RF006','REC0015','Not Available on the booked dates','2021-11-28','Applied for Refund'),
('CAN0020','L0016','G0020','H0016','RF0020','REC0017','Change in travel plans','2019-10-03','Applied for Refund'),
('CAN0013','L0011','G0013','H0011','RF0013','REC0019','Unexpected Weather Events','2025-03-03','Applied for Refund'),
('CAN008','L002','G008','H002','RF008','REC0021','Security reasons flagged by Airbnb[Id Verification Not Completed]','2021-12-24',
'Transaction Cancelled'),
('CAN0019','L0019','G0019','H0019','RF0019','REC0024','Found a Cheaper place','2021-12-17','Applied for Refund'),
('CAN0015','L001','G0015','H001','RF0015','REC0026','Other, Personal Reason','2018-11-25','Applied for Refund'),
('CAN0016','L0015','G0016','H0015','RF0016','REC0028','Payment Failure','2023-09-30','Transaction Failed'),
('CAN0013','L006','G0013','H006','RF0013','REC0030','Change in number of guests','2024-04-17','Help from Customer Service'),
('CAN0014','L0014','G001','H0014','RF001','REC0032','Cancellation made by host: Double Booking or Calendar Error','2019-10-21',
'Providing a Full Refund'),
('CAN0016','L0016','G0010','H0016','RF0010','REC0033','Security reasons flagged by Airbnb[Id Verification Not Completed]',
'2019-08-24','Transaction Cancelled'),
('CAN0012','L0012','G0011','H0012','RF0011','REC0035','Visa Issues','2023-12-18','Help from Customer Service'),
('CAN007','L0013','G007','H0013','RF007','REC0036','Payment Failure','2023-09-30','Transaction Failed'),
('CAN004','L0014','G004','H0014','RF004','REC0038','Found a Cheaper place','2022-05-25','Applied for a Refund'),
('CAN003','L0020','G003','H0020','RF003','REC0039','Security reasons flagged by Airbnb[Id Verification Not Completed]',
'2019-01-13','Transaction Cancelled');
```

Refund:

Create Command:

```
CREATE TABLE IF NOT EXISTS refund(
    refund_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    guest_id character varying(10) NOT NULL,
    host_id character varying(10) NOT NULL,
    cancel_id character varying(10) NOT NULL,
    receipt_id character varying(10) NOT NULL,
    refund_date date NOT NULL,
    refund_status character varying NOT NULL
);
```

Changes in the schema → Primary key: removed

- The refund table stores and manages transactions, often due to cancellation of a booking.
- It helps in processing standard measures for handling refund transactions.

Alter Command:

```
ALTER TABLE IF EXISTS refund
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS refund
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS refund
ADD CONSTRAINT host_id FOREIGN KEY (host_id)
REFERENCES host (host_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS refund
ADD CONSTRAINT cancel_id FOREIGN KEY (cancel_id)
REFERENCES cancellation (cancel_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS refund
ADD CONSTRAINT receipt_id FOREIGN KEY (receipt_id)
REFERENCES receipt (receipt_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Refund: Insert command

Below is the insert command for the refund table, [20 in total]:

```
INSERT INTO refund
VALUES('RF0010','L001','G0010','H001','CAN0010','REC002','2022-10-06','No Refund Applicable'),
('RF0014','L0017','G0014','H0017','CAN0014','REC005','2024-06-07','Partial Refund Processing'),
('RF0016','L0018','G0016','H0018','CAN0016','REC006','2025-03-23','Refund Not Applied'),
('RF006','L0019','G006','H0019','CAN0019','REC006','2023-07-29','Full Refund Initiated'),
('RF004','L006','G004','H006','CAN004','REC0011','2024-09-23','No Refund Applicable'),
('RF001','L0015','G001','H0015','CAN001','REC0013','2024-01-19','No Refund Applicable'),
('RF006','L008','G006','H008','CAN006','REC0015','2021-11-30','Partial Refund Initiated'),
('RF0020','L0016','G0020','H0016','CAN0020','REC0017','2019-10-03','Partial Refund Initiated'),
('RF0013','L0011','G0013','H0011','CAN0013','REC0019','2025-03-08','Full Refund Initiated'),
('REC0021','L002','G008','H002','CAN008','REC0021','2021-12-24', 'No Refund Applicable'),
('RF0019','L0019','G0019','H0019','CAN0019','REC0024','2021-12-19','Partial Refund Applicable'),
('RF0015','L001','G0015','H001','CAN0015','REC0026','2018-11-29','Partial Refund Processing'),
('RF0016','L0015','G0016','H0015','CAN0016','REC0028','2023-09-30','No Refund Applicable'),
('RF0013','L006','G0013','H006','CAN0013','REC0030','2024-04-21','Refund Not Applied'),
('RF001','L0014','G001','H0014','CAN0014','REC0032','2019-10-22','Providing a Full Refund'),
('RF0010','L0016','G0010','H0016','CAN0016','REC0033','2019-08-24','No Refund Applicable'),
('RF0011','L0012','G0011','H0012','CAN0012','REC0035','2023-12-23','Partial Refund Processing'),
('RF007','L0013','G007','H0013','CAN007','REC0036','2023-09-30','No Refund Applicable'),
('RF004','L0014','G004','H0014','CAN004','REC0038','2022-05-31','Partial Refund Initiated'),
('RF003','L0020','G003','H0020','CAN003','REC0039','2019-01-13','No Refund Applicable');
```

Wishlist:

Create Command:

```
CREATE TABLE IF NOT EXISTS wishlists(
    wishlist_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    guest_id character varying(10) NOT NULL,
    wishlist_name character(100) NOT NULL,
    PRIMARY KEY (wishlist_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS wishlists
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS wishlists
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Insert Command:

The wishlist table stores all the wishlists that are created by guests. Below is the insert command for the wishlist table, [20 in total]:

```
INSERT INTO wishlists
VALUES('WL001','L0020','G001','Wishlist_1'),
('WL002','L0019','G002','Wishlist_2'),
('WL003','L0018','G003','Wishlist_3'),
('WL004','L0017','G004','Wishlist_4'),
('WL005','L0016','G005','Wishlist_5'),
('WL006','L0015','G006','Wishlist_6'),
('WL007','L0014','G007','Wishlist_7'),
('WL008','L0013','G008','Wishlist_8'),
('WL009','L0012','G009','Wishlist_9'),
('WL0010','L0011','G0010','Wishlist_10'),
('WL0011','L0010','G0011','Wishlist_11'),
('WL0012','L009','G0012','Wishlist_12'),
('WL0013','L008','G0013','Wishlist_13'),
('WL0014','L007','G0014','Wishlist_14'),
('WL0015','L006','G0015','Wishlist_15'),
('WL0016','L005','G0016','Wishlist_16'),
('WL0017','L004','G0017','Wishlist_17'),
('WL0018','L003','G0018','Wishlist_18'),
('WL0019','L002','G0019','Wishlist_19'),
('WL0020','L001','G0020','Wishlist_20');
```

Region

Create Command:

```
CREATE TABLE IF NOT EXISTS region(
    region_id character varying(10) NOT NULL,
    region_name character varying(100) NOT NULL,
    PRIMARY KEY (region_id)
);
```

Insert Command:

```
INSERT INTO region
VALUES('REG001', 'Africa'),
('REG002', 'Americas'),
('REG003', 'Asia'),
('REG004', 'Europe'),
('REG005', 'Oceania'),
('REG006', 'Polar');
```

- Region table stores geographic allotment and representation of which regions the listings, guests, and hosts are located.

Country:

Create Command:

```
CREATE TABLE IF NOT EXISTS country(
    country_id character varying(10) NOT NULL,
    region_id character varying(10) NOT NULL,
    country_name character(100) NOT NULL,
    PRIMARY KEY (country_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS country
ADD CONSTRAINT region_id FOREIGN KEY (region_id)
REFERENCES region (region_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID;
```

- Country table stores geographic allotment and representation of which countries the listings, guests, and hosts are located.
- Below is the insert command for the country table, only the first statement is expanded here as, every insert statement[55 in total] follows a similar structure.

('C001','REG005','Australia')

Insert Command:

```
INSERT INTO country
VALUES('C001','REG005','Australia'),
('C002','REG004','Austria'),
('C003','REG003','Bahrain'),
('C004','REG003','China'),
('C005','REG003','Cyprus'),
('C006','REG004','Czech Republic'),
('C007','REG004','Denmark'),
('C008','REG003','Egypt'),
('C009','REG005','Fiji'),
('C0010','REG004','Finland'),
('C0011','REG004','France'),
('C0012','REG005','French Polynesia'),
('C0013','REG004','Germany'),
('C0014','REG004','Greece'),
('C0015','REG003','India'),
('C0016','REG004','Italy'),
('C0017','REG003','Iran'),
('C0018','REG004','Ireland'),
('C0019','REG003','Japan'),
('C0020','REG003','Jordan'),
('C0021','REG001','Kenya'),
('C0022','REG003','Kuwait'),
('C0023','REG003','Lebanon'),
('C0024','REG003','Malaysia'),
('C0025','REG001','Mauritius'),
('C0026','REG001','Morocco'),
('C0027','REG001','Nigera'),
('C0028','REG004','Netherlands'),
('C0029','REG005','New Zealand'),
```

City:

Create Command:

```
CREATE TABLE IF NOT EXISTS city(
    city_id character varying(10) NOT NULL,
    country_id character varying(10) NOT NULL,
    city_name character(100) NOT NULL,
    PRIMARY KEY (city_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS city
ADD CONSTRAINT country_id FOREIGN KEY (country_id)
REFERENCES country (country_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID;
```

- City table stores geographic allotment and representation of which countries the listings, guests, and hosts are located.
- Below is the insert command for the city table, only the first statement is expanded here as, every insert statement[78 in total] follows a similar structure.

('COO1','C0043','Woleseley'),

Insert Command:

```
INSERT INTO city
VALUES('COO1','C0043','Woleseley'),
('COO2','C0044','Sindun-myeon'),
('COO3','C0045','Brålanda'),
('COO4','C0013','Göppingen'),
('COO5','C0030','Raelingen'),
('COO6','C0014','Pyrgos Kallistis'),
('COO7','C002','Deutschlandsberg'),
('COO8','C001','Forth'),
('COO9','C0053','Los Angeles'),
('COO10','C0015','Manali'),
('COO11','C0048','Doi Suthep-Pui'),
('COO12','C0010','Ounasvaara'),
('COO13','C0020','Wadi Rum'),
('COO14','C0055','Hạ Long'),
('COO15','C0050','Ortahisa'),
('COO16','C0052','Diggle'),
('COO17','C0019','Sellano'),
('COO18','C0016','Umbria'),
('COO19','C0025','Roches Noires'),
('COO20','C0026','Fès'),
('COO21','C0045','Stockholm'),
('COO22','C0052','Manchester'),
('COO23','C0015','Bengaluru'),
('COO24','C0021','Nairobi'),
('COO25','C00015','Hyderabad'),
('COO26','C0042','Madrid'),
('COO27','C0052','Glasgow'),
('COO28','C0052','New York'),
('COO29','C0024','Kuala Lumpur'),
```

Location:

Create Command:

```
CREATE TABLE IF NOT EXISTS location(
    location_id character varying(10) NOT NULL,
    neighborhood_id character varying(10) NOT NULL,
    location_name character(100) NOT NULL,
    PRIMARY KEY (location_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS location
ADD CONSTRAINT neighboorhood_id FOREIGN KEY (neighborhood_id)
REFERENCES neighborhood (neighborhood_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID;
```

- Location table stores geographic allotment and representation of which countries the listings, guests, and hosts are located.
- Below is the insert command for the location table, only the first statement is expanded here as, every insert statement[80 in total] follows a similar structure.

('LO001','NG001','Western Cape')

Insert Command:

```
INSERT INTO location
VALUES('LO001','NG001','Western Cape'),
('LO002','NG002','Icheon-si'),
('LO003','NG003','Västra Götalands län'),
('LO004','NG004','Baden-Württemberg'),
('LO005','NG005','Akershus'),
('LO006','NG006','Santorini'),
('LO007','NG007','Steiermark'),
('LO008','NG008','Tasmania'),
('LO009','NG009','California'),
('LO0010','NG0010','Himachal Pradesh'),
('LO0011','NG0011','Chiang Mai'),
('LO0012','NG0012','Rovaniemi'),
('LO0013','NG0013','Aqaba Governorate'),
('LO0014','NG0014','Quảng Ninh'),
('LO0015','NG0015','Nevsehir'),
('LO0016','NG0016','England'),
('LO0017','NG0017','Shimane Prefecture'),
('LO0018','NG0018','Umbria'),
('LO0019','NG0019','Rivière du Rempart'),
('LO0020','NG0020','Fès-Meknès'),
('LO0021','NG0021','Kungsbron'),
('LO0022','NG0022','Cross Street'),
('LO0023','NG0023','Gandhi Nagar'),
('LO0024','NG0024','Bernard'),
('LO0025','NG0025','Darus Salam'),
('LO0026','NG0026','Calle Del Olmo'),
('LO0027','NG0027','Monteith Row'),
('LO0028','NG0028','Soho'),
('LO0029','NG0029','Lorong Mayang'),
```

Neighborhood:

Create Command:

```
CREATE TABLE IF NOT EXISTS neighborhood(
    neighborhood_id character varying(10) NOT NULL,
    city_id character varying(10) NOT NULL,
    neighborhood_name character(100) NOT NULL,
    PRIMARY KEY (neighborhood_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS neighborhood
ADD CONSTRAINT city_id FOREIGN KEY (city_id)
REFERENCES city (city_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID;
```

- Neighborhood table stores geographic allotment and representation of which countries the listings, guests, and hosts are located.
- Below is the insert command for the neighborhood table, only the first statement is expanded here as, every insert statement[79 in total] follows a similar structure.

('NG001','COO1','Romansrivier')

Insert Command:

```
INSERT INTO neighborhood
VALUES('NG001','COO1','Romansrivier'),
('NG002','COO2','Sunam-RI'),
('NG003','COO3','Sörbo'),
('NG004','COO4','Stauferpark'),
('NG005','COO5','Smestad'),
('NG006','COO6','Artemis Karamolegos'),
('NG007','COO7','Klein-Preding'),
('NG008','COO8','Melrose'),
('NG009','COO9','Downtown-Los Angeles'),
('NG0010','COO10','Jagatsukh'),
('NG0011','COO11','Rama-Lana'),
('NG0012','COO12','Vesalanperä'),
('NG0013','COO13','Rashid Canyon'),
('NG0014','COO14','Bãi Tâm'),
('NG0015','COO15','Hibe Dede Cd'),
('NG0016','COO16','Delph'),
('NG0017','COO17','Inahara'),
('NG0018','COO18','Molini di Orsano'),
('NG0019','COO19','Plaine des Roches'),
('NG0020','COO20','Andalous'),
('NG0021','COO21','Norrmalm'),
('NG0022','COO22','Hulme'),
('NG0023','COO23','Halasuru'),
('NG0024','COO24','Kilimani'),
('NG0025','COO25','Banjara Hills'),
('NG0026','COO26','Getafe'),
('NG0027','COO27','Govanhill'),
('NG0028','COO28','Union City'),
('NG0029','COO29','Hulu Langat'),
```

Message:

Create Command:

```
CREATE TABLE IF NOT EXISTS msg(
    msg_id character varying(10) NOT NULL,
    msg_text text NOT NULL,
    msg_time time without time zone NOT NULL
);
```

- The Messages table stores communication exchanged between users, typically between guests and hosts
- Below is the insert command for the message table, only the first statement is expanded here as, every insert statement
- [92 in total] follows a similar structure.

Insert Command:

```
INSERT INTO msg
VALUES('MSG00h1','Hello, Thank you for booking your stay with us. We are excited to welcome you and feel free to ask any questions you may have!','21:18')
('MSG00g1','Thank you for the message. Could you please confirm the check-in time?','13:02'),
('MSG00h1','Of course! the earliest check-in time is from 2:00 pm but, if you need to arrive earlier, just let me know and I will do my best to accommodat
('MSG00g1','Yeah sure. I will let you know if that is the case.','01:45'),
('MSG00h2','Hello, thank you so much for booking with us. Is there anything I can help you with?','00:23'),
('MSG00g7','Hi, So it was mentioned in the description that there is someone who also teaches to play a music intsrument of our choice. Is this still vali
('MSG00h2','Hi, yeah, it is still valid although when you check-in, there is a list of instruments that you can select from that is included with listing-
('MSG00g7','Okay, thank you! I will contact you if there is anything else that is needed.','05:27'),
('MSG00h4','Hi, is there anything I can help you with? Please let know','18:04'),
('MSG00g18','Hello, I am actually an art student, preparing for finals and was looking for an inspiration. Can I use your place as an inspiration?','09:16
('MSG00h4','Hello, I am so excited for this. Yes you may use this for your inspiration. All the very best for your final. Can I ask for a favor though?','(
('MSG00g18','Yes please','22:22'),
('MSG00h4','If possible, can we have have painting once it is done and possible for you to give us?. We can pay you for the painting.','20:17'),
('MSG00g18','Yes ofcourse','15:15'),
('MSG00h7','Hi, thank you for booking with us, we are looking forword to host you!','13:09'),
('MSG00g14','Hi, thank you. We have some requests as this is a post-wedding celebration!','23:45'),
('MSG00h7','Hello, many congrats! how can we help?','15:29'),
('MSG00g14','Hi, we just require one honeymoon suite and an open space where we can dance?','16:01'),
('MSG00h7','Yes sure, that would be done the materials used will be charged though, is it okay?','06:23'),
('MSG00g14','yes that is okay. let us know if there is anything else','12:49'),
('MSG00h7','okay, congrats once again!','01:17'),
('MSG00h8','Hello, thank you for booking with us, is there anything we can do to help?','17:01'),
('MSG00g12','Hi, there are no requests from us, but thanks for asking though!','20:23'),
('MSG00h9','Hello, Is there anything special you are hoping from us to do?','13:15'),
('MSG00g2','Hello, since this is a business trip, there are no requests from me, thank you so much for asking!','05:21'),
('MSG00h10','Hi, thank you for selecting to stay with us! Are there any questions for us? feel free to reach out!','16:05'),
('MSG00g14','Greetings, thank you dor reaching out. I do have some questions. I actually want to cancel my booking as something came up that cannot be re-
('MSG00h10','Hello, thank you for reaching out. Is there any possibility that you can re-schedule from current dates to another dates?','00:57'),
```

('MSG00h1','Hello, Thank you for booking your stay with us. We are excited to welcome you and feel free to ask any questions you may have!','21:18')

Changes in the schema → msg_text bytea changed to msg_text text , Primary key, msg_date, host_id and guest_id removed

Review:

Create Command:

```
CREATE TABLE IF NOT EXISTS review(
    review_id character varying(10) NOT NULL,
    guest_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    review_added text NOT NULL,
    review_date date NOT NULL
);
```

Alter Command:

```
ALTER TABLE IF EXISTS review
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS review
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Insert Command:

```
INSERT INTO review
VALUES('REID001','G001','L008','What a beautiful relaxing getaway! Spent our honeymo
('REID0012','G0012','LOO3','This was such an incredible house to stay in. It is a be
('REID0019','G0019','L0020','Beautiful ryad. The host was a courteous professional,
('REID0018','G0018','L004','Absolute dream! Anyone who has the opportunity should bo
('REID007','G007','L002','I really used it comfortably! It was equipped with towels
('REID003','G003','LOO13','The people of the Wadi Rum desert were so very warm and v
('REID002','G002','L009','An amazing place in beautiful surroundings. Breakfast and
('REID0016','G0016','L0013','If you're looking for wadi rum dessert, this is the pla
('REID009','G009','L0010','Firstly the apartment is lovely with all the amenities or
('REID009','G009','L0012','Photos don't do this beautiful alpine cottage justice! Su
('REID0018','G0018','L001','BETTER THAN ADVERTISED! The photos really don''t do it j
('REID003','G003','L0017','The location is perfectly situated nearby the train stat
('REID002','G002','L0018','A unique experience, to do at least once in your life. Th
('REID0017','G0017','L0020','Dar Victoria is beautiful. The staff are VERY accommoda
('REID0020','G0020','L003','Absolutely incredible place to visit and stay! Anders wa
('REID0014','G0014','L007','Warm welcome and fantastic location with beautiful swimm
('REID0014','G0014','L0010','Place was clean, tidy and appreciated the doorman for s
('REID0012','G0012','L008','What a little piece of paradise! The best B & B we''ve e
('REID008','G008','L009','We really loved Ali''s place! The space was beautiful and
('REID001','G001','L001','What an amazing place! This is a part where the place exce
```

Changes made in the schema → Primary Key:removed. rating_added bytea changed to rating_added text for easier output readability

- The review table stores feedback left by a guest about a particular property [listing or room] they have stayed at.
- Below is the insert command for the review table, only the first statement is expanded here as, every insert statement[20 in total] follows a similar structure.

('REID001','G001','L008','What a beautiful relaxing getaway! Spent our honeymoon here and it was exactly what we were looking for. The beauty was unmatched and the privacy was on another level. I will say, try to come in a car that can handle climbing up a dirt road.','2021-12-01')

Ratings

Create Command:

```
CREATE TABLE IF NOT EXISTS rating(
    rating_id character varying(10) NOT NULL,
    guest_id character varying(10) NOT NULL,
    listing_id character varying(10) NOT NULL,
    rating_category character varying(20) NOT NULL,
    overall_rating numeric(5) NOT NULL,
    rating_date date NOT NULL
);
```

Alter Command:

```
ALTER TABLE IF EXISTS rating
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS rating
ADD CONSTRAINT listing_id FOREIGN KEY (listing_id)
REFERENCES listing (listing_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS rating
ADD CONSTRAINT rating_category FOREIGN KEY (rating_category)
REFERENCES rating_category (rating_category_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS rating
ADD CONSTRAINT rating_number FOREIGN KEY (rating_category)
REFERENCES rating_number (rating_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Insert Command:

- The rating table stores score given by users to a listing and host based on their experience while staying at booked property

```
INSERT INTO rating
VALUES('RAID001','G001','L008','RCA1:RA1,RCA2:RA1,RCA3:RA1,RCA4:RA1,RCA5:RA1,RCA6:RA1',4.99,'2021-12-01'),
('RAID0012','G0012','LOO3','RCA1:RA1,RCA2:RA1,RCA3:RA1,RCA4:RA1,RCA5:RA3,RCA6:RA3',4.96,'2021-01-04'),
('RAID0019','G0019','L0020','RCA1:RA1,RCA2:RA1,RCA3:RA1,RCA4:RA3,RCA5:RA3,RCA6:RA1',4.8,'2022-01-03'),
('RAID0018','G0018','L004','RCA1:RA2,RCA2:RA1,RCA3:RA1,RCA4:RA1,RCA5:RA3,RCA6:RA3',5.0,'2022-06-10'),
('RAID007','G007','L002','RCA1:RA4,RCA2:RA2,RCA3:RA1,RCA4:RA2,RCA5:RA2,RCA6:RA3',4.84,'2024-11-13'),
('RAID003','G003','LOO13','RCA1:RA4,RCA2:RA2,RCA3:RA2,RCA4:RA3,RCA5:RCA5:RA2,RCA6:RA3',4.86,'2025-05-02'),
('RAID002','G002','L009','RCA1:RA1,RCA2:RA1,RCA3:RA2,RCA4:RA1,RCA5:RA3,RCA6:RA2',4.95,'2023-06-30'),
('RAID0016','G0016','L0013','RCA1:RA4,RCA2:RA2,RCA3:RA2,RCA4:RA3,RCA5:RA2,RCA6:RA3',4.86,'2023-10-02'),
('RAID009','G009','L0010','RCA1:RA2,RCA2:RA2,RCA3:RA3,RCA4:RA1,RCA5:RA4,RCA6:RA3',4.93,'2019-03-23'),
('RAID009','G009','L0012','RCA1:RA2,RCA2:RA1,RCA3:RA1,RCA4:RA1,RCA5:RA1,RCA6:RA2',4.96,'2023-08-07'),
('RAID0018','G0018','L001','RCA1:RA1,RCA2:RA1,RCA3:RA2,RCA4:RA2,RCA5:RCA5:RA2,RCA6:RA2',4.94,'2024-12-02'),
('RAID003','G003','L0017','RCA1:RA3,RCA2:RA2,RCA3:RA1,RCA4:RCA4:RA1,RCA5:RA1,RCA6:RA4',4.74,'2020-11-05'),
('RAID002','G002','L0018','RCA1:RA1,RCA2:RA2,RCA3:RA2,RCA4:RA1,RCA5:RA2,RCA6:RA2.',5.0,'2024-06-27'),
('RAID0017','G0017','L0020','RCA1:RA1,RCA2:RA1,RCA3:RA1,RCA4:RA3,RCA5:RA3,RCA6:RA1',4.8,'2021-11-29'),
('RAID0020','G0020','L003','RCA1:RA1,RCA2:RA1,RCA3:RA1,RCA4:RA1,RCA5:RA3,RCA6:RA3',4.96,'2024-10-03'),
('RAID0014','G0014','L007','RCA1:RA4,RCA2:RA3,RCA3:RA4,RCA4:RA3,RCA5:RA4,RCA6:R6',4.0,'2023-10-20'),
('RAID0014','G0014','L0010','RCA1:RA2,RCA2:RA2,RCA3:RA3,RCA4:RA1,RCA5:RA4,RCA6:RA3',4.93,'2019-10-15'),
('RAID0012','G0012','L008','RCA1:RA1,RCA2:RA1,RCA3:RA1,RCA4:RA1,RCA5:RA1,RCA6:RA1',4.99,'2021-06-20'),
('RAID008','G008','L009','RCA1:RA1,RCA2:RA1,RCA3:RA2,RCA4:RA1,RCA5:RA3,RCA6:RA2',4.95,'2021-11-29'),
('RAID001','G001','L001','RCA1:RA1,RCA2:RA1,RCA3:RA2,RCA4:RA2,RCA5:RA2,RCA6:RA2',4.94,'2021-12-18');
```

Changes made in the schema → Primary Key: removed. rating_category character varying(10) changed to character varying (50) to accommodate the different categories of ratings

Rating Category

Create Command:

```
CREATE TABLE IF NOT EXISTS rating_category(
    rating_category_id character varying(10) NOT NULL,
    category_name character(100) NOT NULL,
    PRIMARY KEY (rating_category_id)
);
```

Insert Command:

```
INSERT INTO rating_category
VALUES('RC1','Cleanliness'),
('RC2','Accuracy'),
('RC3','Check-in'),
('RC4','Communication'),
('RC5','Location'),
('RC6','Value');
```

Changes made in the schema → Changed category_name character varying (10) to character varying (20) ,category_rating changed to double precision from character varying(100)

- The rating category stores specific aspects of a guest's experience that are rated separately to provide detailed feedback about a listing and host
- This is a reference table for rating

Rating Number

Create Command:

```
CREATE TABLE IF NOT EXISTS rating_number(
    rating_id character varying(10) NOT NULL,
    rating_number double precision NOT NULL
);
```

Insert Command:

```
INSERT INTO
rating_number
VALUES('RA1',5.0),
('RA2',4.9),
('RA3',4.8),
('RA4',4.7),
('RA5',4.6),
('RA6',4.5),
('RA7',4.4),
('RA8',4.3),
('RA9',4.2),
('RA10',4.1),
('RB1',4.0),
('RB2',3.9),
('RB3',3.8),
('RB4',3.7),
('RB5',3.6),
('RB6',3.5),
('RB7',3.4),
('RB8',3.3),
```

Changes in the schema → New table rating_number created

- The rating number stores specific aspects in numerical format of a guest's experience that are rated to provide feedback about a listing and host
- This is a reference table for rating
- Below is the insert command for the rating number table, only the first statement is expanded here as, every insert statement[51 in total] follows a similar structure.

('RA1',5.0)

Pay Category

Create Command:

```
CREATE TABLE IF NOT EXISTS pay_category(
    pay_category_id character varying(10) NOT NULL,
    pay_category character(100) NOT NULL
);
```

Changes in the schema → New table pay_category created

Insert Command:

```
INSERT INTO pay_category
VALUES('PC0dc01','Card'),
('PC0cc02','Card'),
('PC0bc03','Bank Account'),
('PC0oa104','PayPal'),
('PC0oa205','Apple Pay'),
('PC0oa306','Booking Credits');
```

- The pay category stores different types of payment-related attributes or transactions.
- This is a reference table for payments, payouts and saved payment methods.

Payments:

Create Command:

```
CREATE TABLE IF NOT EXISTS payments(
    payment_id character varying(10) NOT NULL,
    guest_id character(10) NOT NULL,
    payment_category character varying(20) NOT NULL,
    payment_date date NOT NULL,
    payment_status character varying(100) NOT NULL,
    payment_amt double precision NOT NULL,
    PRIMARY KEY (payment_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS payments
ADD CONSTRAINT guest_id FOREIGN KEY (guest_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS payments
ADD FOREIGN KEY (payment_category)
REFERENCES pay_category (pay_category_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Changes in the schema →
Changed payment_category character varying (10) to character varying (20)

Insert Command:

```
INSERT INTO payments
VALUES('PY001','Credit Card','G001','2021-11-26','Successful',1558),
('PY002','PayPal','G0010','2022-10-06','Cancelled',1589),
('PY003','Debit Card','G0012','2021-01-02','Successful',1047),
('PY004','Booking Credits','G0019','2021-12-26','Successful',3045),
('PY005','Credit Card','G0014','2024-05-30','Refund',3587),
('PY006','Apple Pay','G0016','2025-03-21','Cancelled',2651),
('PY007','PayPal','G0018','2022-06-06','Successful',1516),
('PY008','Debit Card','G007','2024-10-31','Successful',2817.75),
('PY009','PayPal','G006','2023-07-27','Refund',3734.25),
('PY0010','Credit Card','G003','2025-04-28','Successful',1174.5),
('PY0011','Apple Pay','G004','2024-09-23','Cancelled',1733.6),
('PY0012','Booking Credits','G002','2023-06-27','Successful',913),
('PY0013','PayPal','G001','2024-01-19','Refund',2289),
('PY0014','PayPal','G0016','2023-09-27','Successful',1338.75),
('PY0015','Booking Credits','G006','2021-11-26','Cancelled',1628.55),
('PY0016','Booking Credits','G009','2019-03-13','Successful',4764.6),
('PY0017','Credit Card','G0020','2019-10-01','Cancelled',656.2),
('PY0018','Apple Pay','G009','2023-05-27','Successful',4531.5),
('PY0019','Credit Card','G0013','2025-02-28','Cancelled',644.3),
('PY0020','CreditCard','G0018','2024-11-30','Successful',917.6),
('PY0021','PayPal','G008','2021-12-24','Cancelled',1800),
('PY0022','Credit Card','G003','2020-10-31','Successful',1401.4),
('PY0023','Debit Card','G002','2024-06-22','Successful',2563.6),
('PY0024','Debit Card','G0019','2021-12-17','Refund',3384.5),
('PY0025','Credit Card','G0017','2021-11-26','Successful',1478.4),
('PY0026','Apple Pay','G0015','2018-11-25','Refund',1611.9),
('PY0027','Apple Pay','G0020','2024-09-26','Successful',2845.8),
('PY0028','Booking Credits','G0016','2023-09-30','Cancelled',2808.65),
('PY0029','Credit Card','G0014','2023-09-27','Successful',5907.5),
```

- The payments table records each payment made by a guest for a booking
- Below is the insert command for the payments table, only the first statement is expanded here as, every insert statement[40 in total] follows a similar structure.

('PY001','Credit Card','G001','2021-11-26','Successful',1558)

Payouts:

Create Command:

```
CREATE TABLE IF NOT EXISTS payouts(
    payout_id character varying(10) NOT NULL,
    payout_category character varying(20) NOT NULL,
    host_id character varying(10) NOT NULL,
    payout_date date NOT NULL,
    payout_status character varying(100) NOT NULL,
    payout_amt double precision NOT NULL,
    taxes_added double precision NOT NULL,
    total_payout_amt double precision NOT NULL,
    PRIMARY KEY (payout_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS payouts
ADD CONSTRAINT host FOREIGN KEY (host_id)
REFERENCES host (host_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS payouts
ADD FOREIGN KEY (payout_category)
REFERENCES pay_category (pay_category_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

Changes in the schema →
Changed payout_category character varying (10) to character varying (20), total_payout_amt added

Insert Command:

```
INSERT INTO payouts
VALUES('PO001','Account Number','H008','2021-11-30','Successful',1558,100,1458),
('PO002','PayPal','H001','2022-10-16','Cancelled',1589,100,1489),
('PO003','Debit Card','H003','2021-01-12','Successful',1047,80,967),
('PO004','UPI','H0020','2021-12-31','Successful',3045,200,2845),
('PO005','Account Number','H0017','2024-06-02','Cancelled',3587,200,3387),
('PO006','Apple Pay','H0018','2025-03-30','Cancelled',2651,150,2501),
('PO007','PayPal','H004','2022-06-10','Successful',1516,100,1416),
('PO008','Debit Card','H002','2024-11-01','Successful',2817.75,150,2667.75),
('PO009','PayPal','H0019','2023-07-27','Cancelled',3734.25,200,3534.25),
('PO0010','Account Number','H0013','2025-04-28','Successful',1174.5,90,1084.5),
('PO0011','Apple Pay','H006','2024-09-25','Cancelled',1733.6,90,1643.6),
('PO0012','Booking Credits','H009','2023-06-29','Successful',913,80,833),
('PO0013','PayPal','H0015','2024-01-20','Cancelled',2289,100,2189),
('PO0014','PayPal','H0013','2023-09-29','Successful',1338.75,95,1243.75),
('PO0015','Booking Credits','H008','2021-11-27','Cancelled',1628.55,85,1543.55),
('PO0016','Booking Credits','H0010','2019-03-13','Successful',4764.6,210,4554.6),
('PO0017','Account Number','H0016','2019-10-03','Cancelled',656.2,85,571.2),
('PO0018','Apple Pay','H0012','2023-05-28','Successful',4531.5,210,4321.5),
('PO0019','Account Number','H0011','2025-02-28','Cancelled',644.3,80,564.3),
('PO0020','Account Number','H001','2024-12-11','Successful',917.6,90,827.6),
('PO0021','PayPal','H002','2021-12-24','Cancelled',1800,100,1700),
('PO0022','Account Number','H0017','2020-10-31','Successful',1401.4,95,1306.4),
('PO0023','Debit Card','H0018','2024-06-26','Successful',2563.6,150,2413.6),
('PO0024','Debit Card','H0019','2021-12-18','Cancelled',3384.5,200,3184.5),
('PO0025','Account Number','H0020','2021-11-27','Successful',1478.4,100,1378.4),
('PO0026','Apple Pay','H001','2018-11-25','Cancelled',1611.9,90,1521.9),
('PO0027','Apple Pay','H003','2024-09-26','Successful',2845.8,210,2635.8),
('PO0028','Booking Credits','H0015','2023-09-30','Cancelled',2808.65,200,2608.65),
('PO0029','Account Number','H007','2023-09-28','Successful',5907.5,300,5607.5),
```

- The net amount that hosts earn from their guests stays after the deduction of service fees, applicable taxes, and any other fees.
- Below is the insert command for the payouts table, only the first statement is expanded here as, every insert statement [40 in total] follows a similar structure.

('PO001','Account Number','H008','2021-11-30','Successful',1558,100,1458)

Saved Payment Methods

Create Command:

```
CREATE TABLE IF NOT EXISTS saved_payment_methods(
    saved_pay_id character varying(10) NOT NULL,
    saved_pay_category character varying(100) NOT NULL,
    category_name character(100) NOT NULL,
    PRIMARY KEY (saved_pay_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS saved_payment_methods
ADD CONSTRAINT saved_pay_id_for_host FOREIGN KEY (saved_pay_id)
REFERENCES host (host_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS saved_payment_methods
ADD CONSTRAINT saved_pay_id_for_guest FOREIGN KEY (saved_pay_id)
REFERENCES guest (guest_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;

ALTER TABLE IF EXISTS saved_payment_methods
ADD CONSTRAINT pay_id FOREIGN KEY (saved_pay_category)
REFERENCES pay_category (pay_category_id) MATCH SIMPLE
ON UPDATE NO ACTION ON DELETE NO ACTION NOT VALID;
```

- Saved payment details that a user has added to their account for easier and faster transactions when booking accommodations.
- Below is the insert command for the id verification table, only the first statement is expanded here as, every insert statement[60 in total] follows a similar structure.
  ('IDVF00g1','IDC0010')

Insert Command:

```
INSERT INTO saved_payment_methods
VALUES('SPMg001','PC0cc02','Credit Card'),
('SPMg002','PC0dc01','Debit Card'),
('SPMg003','PC0oa306','Booking Credits'),
('SPMg004','PC0oa104','PayPal'),
('SPMg005','PC0dc01','Debit Card'),
('SPMg006','PC0oa306','Booking Credits'),
('SPMg007','PC0oa306','Booking Credits'),
('SPMg008','PC0oa104','PayPal'),
('SPMg009','PC0oa306','Booking Credits'),
('SPMg0010','PC0oa205','Apple Pay'),
('SPMg0011','PC0oa306','Booking Credits'),
('SPMg0012','PC0cc02','Credit Card'),
('SPMg0013','PC0dc01','Debit Card'),
('SPMg0014','PC0oa306','Booking Credits'),
('SPMg0015','PC0oa205','Apple Pay'),
('SPMg0016','PC0oa306','Booking Credits'),
('SPMg0017','PC0oa306','Booking Credits'),
('SPMg0018','PC0dc01','Debit Card'),
('SPMg0019','PC0oa306','Booking Credits'),
('SPMg0020','PC0dc01','Debit Card'),
('SPMh001','PC0bc03','Account Number'),
('SPMh002','PC0dc01','Debit Card'),
('SPMh003','PC0cc02','Credit Card'),
('SPMh004','PC0oa104','PayPal'),
```

Changes in the schema → saved_pay_category changed from bytea to character varying(100)

ID Category

Create Command:

```
CREATE TABLE IF NOT EXISTS id_category(
    id_category_id character varying(10) NOT NULL,
    id_category character varying(100) NOT NULL
);
```

Insert Command:

```
INSERT INTO id_category
VALUES('IDC001','Email','Not Verified'),
('IDC002','Phone Number','Not Verified'),
('IDC003','Govt. Recognised Id','Not Verified'),
('IDC004','Email','In Process'),
('IDC005','Phone Number','In Process'),
('IDC006','Govt, Recognised Id','In Process'),
('IDC007','Email','Verified'),
('IDC008','Phone Number','Verified'),
('IDC009','Govt. Recognised Id','Verified'),
('IDC0010','All IDs','Verified');
```

- The id category stores different types of Id verification status to check if guest, host and admin are verified.
- This is a reference table for id verification

Id Verification:

Create Command:

```
CREATE TABLE IF NOT EXISTS id_verification(
    id_verify character varying(10) NOT NULL,
    id_type_status character(100) NOT NULL,
    PRIMARY KEY (id_verify)
);
```

Changes made in the schema → cols deleted: host_id, guest_id, verify_date; renamed verify_id to id_verify

Alter Command:

```
ALTER TABLE IF EXISTS id_verification
    ADD FOREIGN KEY (id_type_status)
    REFERENCES id_category (id_category_id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE NO ACTION
    NOT VALID;
```

- ID verification table stores and securely processes users identification data to confirm their identity before allowing them to participate for booking or hosting a listing
- Below is the insert command for the id verification table, only the first statement is expanded here as, every insert statement[60 in total] follows a similar structure.

('IDVF00g1','IDC0010')

Insert Command:

```
INSERT INTO id_verification
VALUES('IDVF00g1','IDC0010'),
('IDVF00g2','IDC0010'),
('IDVF00g3','IDC001'),
('IDVF00g4','IDC004'),
('IDVF00g5','IDC0010'),
('IDVF00g6','IDC0010'),
('IDVF00g7','IDC0010'),
('IDVF00g8','IDC002'),
('IDVF00g9','IDC0010'),
('IDVF0g10','IDC003'),
('IDVF0g11','IDC0010'),
('IDVF0g12','IDC0010'),
('IDVF0g13','IDC0010'),
('IDVF0g14','IDC0010'),
('IDVF0g15','IDC0010'),
('IDVF0g16','IDC0010'),
('IDVF0g17','IDC0010'),
('IDVF0g18','IDC0010'),
('IDVF0g19','IDC0010'),
('IDVF0g20','IDC0010'),
('IDVF00h1','IDC005'),
('IDVF00h2','IDC001'),
('IDVF00h3','IDC0010'),
('IDVF00h4','IDC0010'),
('IDVF00h5','IDC0010'),
('IDVF00h6','IDC0010'),
```

Coupon

Create Command:

```
CREATE TABLE IF NOT EXISTS coupon(
    coupon_id character varying(10) NOT NULL,
    coupon_name character varying(100) NOT NULL,
    coupon_status character varying(100) NOT NULL,
    coupon_validity daterange NOT NULL,
    PRIMARY KEY (coupon_id)
);
```

- A coupon table stores unique promotional code or credit that users can apply to reduce the cost of their bookings.
- These coupons are typically one-time use and have an expiry date.

Insert Command:

```
INSERT INTO coupon
VALUES('CC001', 'New-User[10% Off]','Valid', '(2024-09-09, 2024-10-09)'),
('CC002', 'Mother''s Day[30% Off]','InValid', '(2019-01-01, 2019-03-01)'),
('CC003', 'Father''s Day[30% Off]','Valid', '(2025-01-20, 2025-05-20)'),
('CC004', 'Winter Wonderland[10% Off]','InValid', '(2025-02-19, 2025-02-28)'),
('CC005', 'Missed-You-Guest[15% Off]','Valid', '(2025-04-12, 2025-04-22)'),
('CC006', 'Christmas-Offer[25% Off]','InValid', '(2018-12-23, 2019-01-13)'),
('CC007', 'Easter-Fun[10% Off]','InValid', '(2019-05-13, 2019-05-23)'),
('CC008', 'Summer-with-Sun[10% Off]','Valid', '(2025-04-04, 2025-04-14)'),
('CC009', 'Happy Birthday[35% Off] ','Valid', '(2025-10-01, 2025-10-10)'),
('CC0010', 'Flat-15 % off with Credit-Pay','Valid', '(2025-01-01, 2025-11-01)'),
('CC0011', 'Flat-25 % off with Apple-Pay','Valid', '(2025-01-02, 2025-11-02)'),
('CC0012', 'Flat-15 % off with Booking Credits','Valid', '(2025-01-10, 2025-11-10)'),
('CC0013', 'New-Year-New-You[20% Off]','InValid', '(2021-12-30, 2022-01-09)'),
('CC0014', 'Valentine''s-Day[25% Off]','InValid', '(2023-02-05, 2023-02-15)'),
('CC0015', 'St.Patrick-Day[10% Off]','InValid', '(2022-04-20, 2022-04-30)'),
('CC0016', 'Ramadan Offer[10% Off]','InValid', '(2018-09-09, 2028-10-09)'),
('CC0017', 'Happy-Anniversary[35% Off]','Valid', '(2024-09-09, 2024-10-09)'),
('CC0018', 'Debit Card Offer[15% Off]','Valid', '(2025-11-15, 2025-11-25)'),
('CC0019', 'Party-101[15% Off]','Valid', '(2025-01-01, 2025-11-01)'),
('CC0020', 'Black Friday[20% Off]','Valid', '(2025-01-01, 2025-12-01)');
```

- Here is the insert command for the coupon table, [20 in total]

GiftCard

- A prepaid credit that users can redeem and apply toward bookings.
- These giftcards usually are available in physical format and also available online on the platform.

Create Command:

```
CREATE TABLE IF NOT EXISTS giftcard(
    giftcard_id character varying(10) NOT NULL,
    giftcard_name character varying(100) NOT NULL,
    giftcard_status character varying(100) NOT NULL,
    giftcard_validity daterange NOT NULL,
    PRIMARY KEY (giftcard_id)
);
```

Insert Command:

```
INSERT INTO giftcard
VALUES('GC001','Thank-You[20% Off]','Valid','(2024-09-10,2024-10-09)'),
('GC002','Sweet Sibling[20% Off]','InValid','(2019-01-02,2019-03-01)'),
('GC003','Best Mom[30% Off]','Valid','(2025-01-21,2025-05-20)'),
('GC004','Best Dad[30% Off]','Valid','(2025-02-20,2025-02-28)'),
('GC005','Winter Wonderland[10% Off]','Valid','(2025-04-13,2025-04-22)'),
('GC006','Christmas Celebrations[25% Off]','InValid','(2018-12-24,2019-01-13)'),
('GC007','Entertaining Easter[15% Off]','InValid','(2019-05-14,2019-05-23)'),
('GC008','Summer-with-Sun[10% Off]','Valid','(2025-04-05,2025-04-14)'),
('GC009','Birthday-Bash[35% Off]','Valid','(2025-10-02,2025-10-10)'),
('GC0010','Flat-15 % off with Credit-Pay','Valid','(2025-01-02,2025-11-01)'),
('GC0011','Flat-25 % off with Apple-Pay','Valid','(2025-01-03,2025-11-02)'),
('GC0012','Flat-15 % off with Booking Credits','Valid','(2025-01-11,2025-11-10)'),
('GC0013','New-Year-New-You[20% Off]','InValid','(2021-12-31,2022-01-09)'),
('GC0014','Valentine''s-Day[25% Off]','InValid','(2023-02-06,2023-02-15)'),
('GC0015','St.Patrick-Day[10% Off]','InValid','(2022-04-21,2022-04-30)'),
('GC0016','Ramadan Offer[10% Off]','InValid','(2018-09-10,2018-10-09)'),
('GC0017','Happy-Anniversary[35% Off]','Valid','(2024-09-10,2024-10-09)'),
('GC0018','Black Friday[20% Off]','Valid','(2025-11-16,2025-11-25)'),
('GC0019','Graduation-Celebration[15% Off]','Valid','(2025-01-02,2025-11-01)'),
('GC0020','BFF[20% Off]','Valid','(2025-01-02,2025-12-01)');
```

- Here is the insert command for the giftcard table, [20 in total]

Guide Book for Guest

Create Command:

```
CREATE TABLE IF NOT EXISTS guide_book_for_guest(
    guide_guest_id character varying(10) NOT NULL,
    guide_name character varying(100) NOT NULL,
    guide_published text NOT NULL,
    guide_category character varying(100) NOT NULL,
    PRIMARY KEY (guide_guest_id)
);
```

Changes made in the schema → guide_publised bytea changed to guide_publised text, guide_name changed to character varying (10)→(100)

Insert Command:

```
INSERT INTO guide_book_for_guest
VALUES('GFG001','Getting Started','Whether a Host, guest, or Airbnb is trying to r
('GFG002','Setting Up Guest Account','Before you can book a reservation a few thin
('GFG003','Know About Your Host','Hosts appreciate guests who treat their place as
('GFG004','Booking for Business,Travel & Events','Arranging accommodation for busi
('GFG005','Seacrh Tips','To make sure the space is just right, use the search filt
('GFG006','FAQ','Need a little more help or have a complaint? Contact us by email,
('GFG007','Commuicate with Host','Once you find a place you like, you may still ha
('GFG008','Check Verified Listing','We require primary hosts, co-hosts and booking
('GFG009','Changes in Resevation','If you sent the host of a place to stay a reser
('GFG0010','Proper First Aid Tips','If you'll be more than an hour away from medic
('GFG0011','Practise Safety Concerns','Safety is one of our top priorities. To hel
('GFG0012','Background Checks','Keeping our Airbnb family safe is one of our top p
('GFG0013','Cancellation Policy/Rules','Cancellation policies are set by the host
('GFG0014','Refund Policy/Rules','You may be entitled to a full refund or one larg
```

- Guidebook for guests table that stores essential information designed to enhance the guest experience during their stay.
- Below is the insert command for the guide book for guest table, only the first statement is expanded here as, every insert statement[14 in total] follows a similar structure.

('GFG001','Getting Started','Whether a Host, guest, or Airbnb is trying to reach you, it's important to keep your account updated. Once you're logged in, go to your Account to edit things like your email address or phone number. ','Help Centre'),

Guide Book By Host:

Create Command:

```
CREATE TABLE IF NOT EXISTS guide_published_by_host(
    published_guide_id character varying(10) NOT NULL,
    host_id character varying(10) NOT NULL,
    guide_name character(100) NOT NULL,
    published_guide text NOT NULL,
    guide_category character varying(100) NOT NULL,
    PRIMARY KEY (published_guide_id)
);
```

Alter Command:

```
ALTER TABLE IF EXISTS guide_published_by_host
    ADD CONSTRAINT host_id FOREIGN KEY (host_id)
    REFERENCES host (host_id) MATCH SIMPLE
    ON UPDATE NO ACTION
    ON DELETE NO ACTION
    NOT VALID;
```

Changes made in the schema → Changed published_guide bytea to published_guide text for easier output readability

Insert Command:

```
INSERT INTO guide_published_by_host
VALUES('PGID001','H001','GuideBook for CapeTown',
('PGID002','H002',null,null,null,null),
('PGID003','H003',null,null,null,null),
('PGID004','H004',null,null,null,null),
('PGID005','H005','What to do in Huser Farm','Loc
('PGID006','H006',null,null,null,null),
('PGID007','H007',null,null,null,null),
('PGID008','H008','Places to Eat and Things to Do
('PGID009','H009','Ali''s Los Angeles GuideBook',
('PGID0010','H0010',null,null,null,null),
('PGID0011','H0011',null,null,null,null),
('PGID0012','H0012',null,null,null,null),
('PGID0013','H0013',null,null,null,null),
('PGID0014','H0014','Hoa''s GuideBook','Seafood
('PGID0015','H0015',null,null,null,null),
('PGID0016','H0016','Nic''s GuideBook','Food Scer
('PGID0017','H0017','Kizuki line switchback throu
('PGID0018','H0018',null,null,null,null),
('PGID0019','H0019',null,null,null,null),
('PGID0020','H0020','comment arrivée chez nous','
```

- This table stores all the guides that are created and published by hosts.
- Here is the insert command for the guide book by host table, only the first statement is expanded here as, every insert statement [20 in total] follows a similar structure.

('PGID001','H001','GuideBook for CapeTown','Cape Town offers a variety of attractions, from iconic landmarks to natural wonders. For a comprehensive experience, consider visiting Table Mountain, V&A Waterfront, Kirstenbosch National Botanical Garden, Robben Island, and Cape Point Nature Reserve.If you''re craving for a great vacation, you should plan a trip to Cape Town, where most of the travellers go in the months of December, January and February. And you will have perfect attractions to explore like Iziko South African Museum.','Travel')

Guide Book for Host

Create Command:

```
CREATE TABLE IF NOT EXISTS guide_book_for_host(
    guide_host_id character varying NOT NULL,
    guide_name character(100) NOT NULL,
    guide_published text NOT NULL,
    guide_category character varying(100) NOT NULL,
    PRIMARY KEY (guide_host_id)
);
```

Insert Command:

```
INSERT INTO guide_book_for_host
VALUES('GFH001','Getting Started','If you’re interested in hosting your space, we’
('GFH002','Setting Up Host Account','Creating an account is free and easy! We’ll n
('GFH003','Creating a Listing','You can create a listing in just a few steps, all
('GFH004','Help with Reservations','Congrats! Someone out there is hoping to call
('GFH005','Sustainability','Booking a listing requires all guests to follow ground
('GFH006','Tools and Features','To help you feel more comfortable hosting, there a
('GFH007','FAQ','Need a little more help or have a complaint? Contact us by email,
('GFH008','Setting Up Property Rules','Once you find a place you like, you may sti
('GFH009','Commmunicate with Guests','Our community is built on the principles of
('GFH0010','Changes in Resevation','If a guest cancels their booking, your cancell
('GFH0011','Proper First Aid Tips','If you’ll be more than an hour away from medic
('GFH0012','Account Management','When you start to host, and periodically througho
('GFH0013','Practise Safety Concerns','Safety is one of our top priorities. To hel
('GFH0014','Background Checks','Keeping our family safe is one of our top prioriti
('GFH0015','Cancellation Policy/Rules','Cancellation policies are set by the host
('GFH0016','Refund Policy/Rules','It depends on how a cancellation would affect yo
('GFH0017','Calendar & Booking Settings','Changing your trip lengths can open your
('GFH0018','Marketing and Promotion','Running a promotion is a great way to stand
('GFH0019','Hosting Resources','Hosting resources or availability settings and pri
```

Changes made in the schema → guide_publised bytea changed to guide_publised text

- Guidebook for Hosts stores guidebooks host can use to learn, organize and improve their services to ensure guests have a smooth, informed, and enjoyable stay
- Below is the insert command for the guide book for host table, only the first statement is expanded here as, every insert statement[19 in total] follows a similar structure.

('GFH001','Getting Started','If you’re interested in hosting your space, we’ll guide you through the process step-by-step. Make sure your home is clean, clutter-free, and that everything is working properly.','Help Centre'),

03

Testing Of Cases

Testing of data using test cases

Test Case 1 & Result

```
1  SELECT * FROM booking
2  WHERE booking_status = 'Cancelled'
3  ORDER BY listing_id;
```

Below is the result for the booking table.
It provides all the details of booking;
where booking status is **CANCELLED**.

- The **SELECT** command is used to select all the details from booking table.
- **WHERE** clause is used for **filtering** the booking status to **cancelled**.
- The **ORDER** query is applied to the result to achieve result in specified order. Here, the **ORDER** of the result in **ASCENDING** as per listing_id.

	booking_id [PK] character	guest_id character	listing_id character	number_of_guests character varying (200)	number_of_pets character varying	nightly_price double precision	staying_for character varying	payable_amt double precision	pay_method character varying (20)	booking_date date	booking_status character varying (100)	security_deposit boolean
1	BO002	G0010	L001	2 Adults	1 Dog	547	3 Nights	1641	PayPal	2022-10-06	Cancelled	false
2	BO0019	G0013	L0011	2 Adults, 2 Children	1 Cat, 1 Dog	341	2 Nights	682	Apple Pay	2025-02-28	Cancelled	true
3	BO0035	G0011	L0012	4 Adults	0	341	4 Nights	1364	PayPal	2023-12-18	Cancelled	true
4	BO0036	G007	L0013	6 Adults	0	445	3 Nights	1335	Credit Card	2023-09-30	Cancelled	true
5	BO0038	G004	L0014	3 Adults	0	780	9 Nights	7020	Credit Card	2022-05-25	Cancelled	true
6	BO0032	G001	L0014	13 Adults	2 Cats	780	12 Nights	9360	Credit Card	2019-10-21	Cancelled	false
7	BO0028	G0016	L0015	2 Adults, 3 Children	0	967	4 Nights	3868	Booking Credits	2023-09-30	Cancelled	true
8	BO0013	G001	L0015	2 Adults	0	967	3 Nights	2901	PayPal	2024-01-19	Cancelled	true
9	BO0017	G0020	L0016	1 Adult	1 Dog	321	2 Nights	642	Credit Card	2019-10-01	Cancelled	false
10	BO0033	G0010	L0016	2 Adults, 1 Child	0	321	2 Nights	642	Debit Card	2019-08-24	Cancelled	false
11	BO005	G0014	L0017	6 Adults	0	562	7 Nights	3934	Credit Card	2024-05-30	Cancelled	true
12	BO006	G0016	L0018	2 Adults, 2 Children	0	624	3 Nights	2496	Apple Pay	2025-03-21	Cancelled	true
13	BO0024	G0019	L0019	3 Adults, 1 Child	1 Parrot	999	6 Nights	5994	Debit Card	2021-12-17	Cancelled	true
14	BO009	G006	L0019	2 Adults	0	999	5 Nights	4995	PayPal	2023-07-27	Cancelled	true
15	BO0021	G008	L002	1 Adult	1 Hamster	655	4 Nights	2620	PayPal	2021-12-24	Cancelled	true
16	BO0039	G003	L0020	2 Adults	1 Cat	731	4 Nights	2924	Apple Pay	2019-01-13	Cancelled	true
17	BO0026	G0015	L005	2 Adults	1 Dog	547	3 Nights	1641	Apple Pay	2018-11-25	Cancelled	false
18	BO0030	G0013	L006	3 Adults	0	433	4 Nights	1732	PayPal	2024-04-17	Cancelled	true
19	BO0011	G004	L006	2 Adults	0	433	4 Nights	1732	Apple Pay	2024-09-23	Cancelled	true
20	BO0015	G006	L008	2 Adults	2 Cats	398	7 Nights	2786	Booking Credits	2021-11-26	Cancelled	false

Test Case 2

```sql
SELECT guest.guest_name, listing.listing_name, host.host_name, booking.booking_id,
receipt.receipt_id, receipt.total_amt, booking.payable_amt, listing.is_security_deposit
receipt.security_deposit_status, booking.pay_method, booking.booking_status
FROM receipt
JOIN booking
    ON receipt.booking_id = booking.booking_id
JOIN listing
    ON listing.listing_id = booking.listing_id
JOIN host
    ON listing.host_id = host.host_id
JOIN guest
    ON booking.guest_id = guest.guest_id
ORDER BY booking.booking_id;
```

- The **SELECT** command here, is used to select the specified columns[guest_name, listing_name, host_name, booking_id, receipt_id, total_amt, payable_amt, is_security_deposit, security_deposit_status, pay_method and booking_status] from tables Guest, Host, Listing, Booking and Receipt.
- **JOIN** clause is used for **joining** one or more columns from different tables together. Here , the booking table is joined to receipt table via the foreign key [booking_id from receipt to booking_id from booking], the listing table is joined to booking table via foreign key [listing_id from booking to listing_id from listing]. Similarly, the host table is joined to listing table via foreign key [host_id from host to host_id from listing] and the guest table is joined to listing table via foreign key [guest_id from guest to host_id from listing]
- The **ORDER** of the result in this case is **ASCENDING** as per booking_id.

Result of TEST CASE 2

Below is the result for test case 2. It provides the selected columns as per listed in select query

	guest_name character (100)	listing_name character (100)	host_name character (100)	booking_id character var	receipt_id character vary	total_amt double prec	payable_amt double precisi	is_security_deposit boolean	security_deposit_status character varying (100)	pay_method character varying (	booking_status character varyin
1	Ahmed Al-Farsi	Cottage: Bed & Breakfast by the River	Benjamin Clark	BO001	REC001	1732	1592	false	Not Applicable	Credit Card	Confirmed
2	Mehmet Yılmaz	Desert EnSuite Double Tents	Ali Karimi	BO0010	REC0010	1305	890	true	Paid	Credit Card	Confirmed
3	Isabelle Dubois	Anemoscope Windmill Villa	Francisco Gomez	BO0011	REC0011	2167	1732	true	Paid	Apple Pay	Cancelled
4	Priya Sharma	Apartment Rental	Siti Nurhaliza	BO0012	REC0012	1218	1098	false	Not Applicable	Booking Credits	Confirmed
5	Ahmed Al-Farsi	Cave House	Olga Ivanova	BO0013	REC0013	3271	2901	true	Paid	PayPal	Cancelled
6	Aisha Al-Thani	Desert EnSuite Double Tents	Ali Karimi	BO0014	REC0014	1785	1335	true	Paid	PayPal	Confirmed
7	Sophie Müller	Cottage: Bed & Breakfast by the River	Benjamin Clark	BO0015	REC0015	2961	2786	false	Not Applicable	Booking Credits	Cancelled
8	Layla Al-Hariri	Dome Stays	William Evans	BO0016	REC0016	5294	4854	true	Paid	Booking Credits	Confirmed
9	Wei Zhang	Tiny Home	Emily Carter	BO0017	REC0017	772	642	false	Not Applicable	Credit Card	Cancelled
10	Layla Al-Hariri	Glass Igloo Stays	Thandiwe Ndlovu	BO0018	REC0018	5035	4550	true	Paid	Debit Card	Confirmed
11	Liam Brown	Leafy Green Earthen Home	Sophie Wilson	BO0019	REC0019	758	682	true	Not Applicable	Apple Pay	Cancelled
12	Henrik Johansson	CouplesPod House Portfolio	Sara Lindholm	BO002	REC002	1766	1641	false	Not Applicable	PayPal	Cancelled
13	Hiroshi Tanaka	CouplesPod House Portfolio	Sara Lindholm	BO0020	REC0020	1147	547	false	Paid	Credit Card	Confirmed
14	Amina Ben Ali	Emotional Healing House	Oliver Smith	BO0021	REC0021	3000	2620	true	Paid	PayPal	Cancelled
15	Mehmet Yılmaz	Naniwa Ryokan	Ahmed Musa	BO0022	REC0022	2156	1686	true	Paid	Credit Card	Confirmed
16	Priya Sharma	TreeHouse Casarampi	Fatima Diallo	BO0023	REC0023	3016	2496	true	Paid	Debit Card	Confirmed
17	Anna Nowak	Island Villa: Entire Unit	Alisha Khan	BO0024	REC0024	6769	5994	true	Paid	Debit Card	Cancelled
18	Dimitri Papadakis	Historical Room Stay	Jessica Taylor	BO0025	REC0025	2112	1462	true	Paid	Credit Card	Confirmed
19	Nikola Petrović	WonderINN Mirrored Glass Cabin	Sunita Reddy	BO0026	REC0026	1791	1641	false	Not Applicable	Apple Pay	Cancelled
20	Wei Zhang	Living Greenhouse Off the Grid	Chinedu Okafor	BO0027	REC0027	3348	3143	false	Not Applicable	Apple Pay	Confirmed

Test Case 3

```sql
SELECT  host_name AS user_name,
host_email AS user_email,
id_verify AS verification_id,
id_status AS id_verification_status
FROM id_verification
JOIN host
    ON host.verify_id = id_verification.id_verify
    JOIN id_category
    ON id_verification.id_type_status = id_category.id_category_id
    WHERE id_verification.id_type_status = 'IDC0010'
UNION
SELECT guest_name, guest_email, id_verify, id_status
FROM id_verification
JOIN guest
    ON guest.verify_id = id_verification.id_verify
    JOIN id_category
    ON id_verification.id_type_status = id_category.id_category_id
    WHERE id_verification.id_type_status = 'IDC0010'
UNION
SELECT admin_name, admin_email, id_verify, id_status
FROM id_verification
JOIN admin
    ON admin.verify_id = id_verification.id_verify
    JOIN id_category
    ON id_verification.id_type_status = id_category.id_category_id
    WHERE id_verification.id_type_status = 'IDC0010'
```

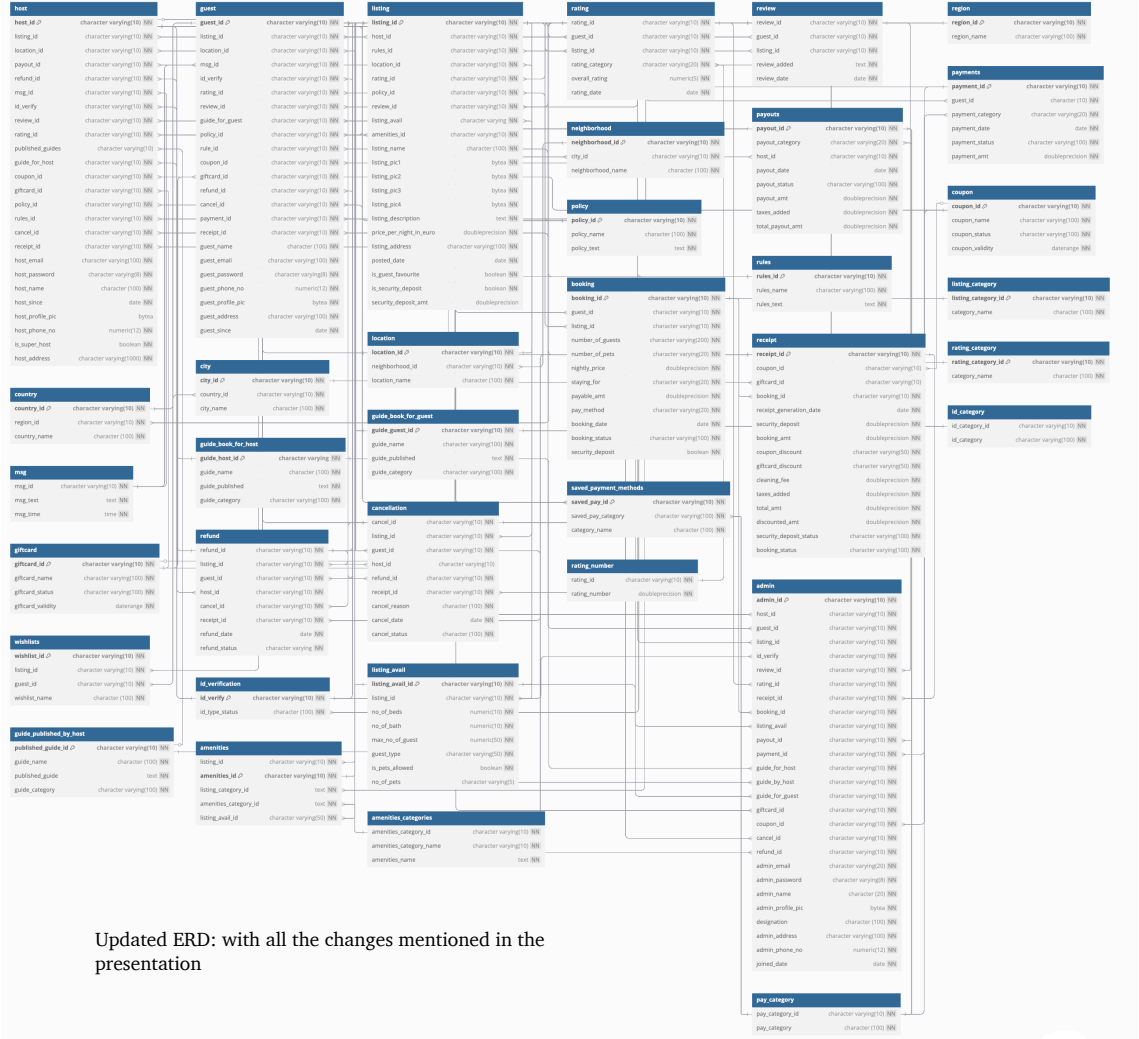
- The **SELECT** command here, is used to select the specified columns[guest_name, guest_email, host_name, host_email, admin_name, admin_email, id_verify, id_status] from tables Guest, Host, Admin and Id Verification.
- **AS** clause is used for renaming columns only in result. Here this was used to rename the selected columns.
- **JOIN** clause here is used to join the id_verification table to host table, guest table and admin table via the foreign key [id_verify from id_verification to host_id from host, guest_id from guest to id_verify from id_verification and admin_id from admin to id_verify from id_verification]. Id Category table is joined to Id Verification table via foreign key [id_type_status from id_verification to id_category_id from id_category for host, guest and admin tables].
- The **UNION** clause is used to join the rows of the table. Here the resulted columns from **JOIN** clause are joined with union to create one table of all the users that are fully verified
- **WHERE** clause is used to filter the id_type_status for id where the id = **IDC0010**; which signifies that the user is fully verified and select those users only

Result of TEST CASE 3

This is the result of test case 3. It provides the selected columns as user_name, user_email, verification_id, id_verification_status. This query returns those users who are fully verified users.

	user_name character	user_email character varying (200)	verification_id character varying (10)	id_verification_status character (100)
1	Anna Nowak	anna.nowak@polmail.pl	IDVF0g19	Verified
2	Fatima Al-Saud	fatima.alsaud@mailnet.sa	IDVF00a2	Verified
3	Siti Nurhaliza	nurhaliza12@gmail.com	IDVF00h9	Verified
4	Francisco Gomez	francisco_g12@mail.com	IDVF00h6	Verified
5	Ahmed Musa	ahmed_2010@mail.ng	IDVF0h17	Verified
6	Nikola Petrović	nikola.petrovic@serbmail.rs	IDVF0g15	Verified
7	Gracie Kim	grace.kim@mail.kr	IDVF00g7	Verified
8	Hannah Turner	hannah.turner@mail.co.uk	IDVF0g14	Verified
9	Sophie Müller	sophie.müeller@mailservice.de	IDVF00g6	Verified
10	Charlotte Moore	charlotte_m@hotmail.com	IDVF00h7	Verified
11	Sophie Wilson	sophie_wilson@mail.co.uk	IDVF0h11	Verified
12	Thandiwe Ndlovu	thandiwe_nn2002@gmail.com	IDVF0h12	Verified
13	Emily Davis	emily.davis@mail.com	IDVF0g11	Verified
14	Ahmed Al-Farsi	ahmed_alFarsi1@postse.se	IDVF00g1	Verified
15	William Evans	will_evans89@gmail.com	IDVF0h10	Verified
16	Mohammed Al-Qasimi	mohammed.alqasimi@gulfmail.bh	IDVF00a6	Verified
17	Chinedu Okafor	chinedu_okafor@gmail.com	IDVF00h3	Verified
18	Benjamin Clark	benjamin_c@gmail.com	IDVF00h8	Verified
19	Fatima Diallo	fatima.d902@outlook.com	IDVF0h18	Verified
20	Lukas Novak	lukas.novak@mailcz.cz	IDVF00a7	Verified

THANK YOU!



Updated ERD: with all the changes mentioned in the presentation